

SELF STORAGE TYCOON

Master Game Design Document

Version 1.1 - Art + Audio Complete Target-Product Specification

Build the property. Run the business. Grow the company.

LAYOUT -> OPERATIONS -> ECONOMICS -> GROWTH

Warm, readable, stylized 3D realism with miniature-diorama clarity.

How to Use This Master GDD

- Parts I-III define the game, game modes, empty-lot start, and the tutorial.
- Parts IV-VIII define construction, interiors, storage products, logistics, utilities, security, and amenities.
- Parts IX-XIII define customers, staff, maintenance, leasing, economics, operating conditions, failure, and diagnostics.
- Parts XIV-XVII define progression, property archetypes, scenarios, sandbox, and the multi-property company layer.
- Part XVIII now contains the full Art Direction Bible and Audio / Sound / Music Direction Bible.
- Parts XIX-XXII define technical authority, save/resume, diagnostics, causal contracts, acceptance tests, production sequencing, scope, and final product vision.

Important: this is the target game specification. Live GitHub remains the authority for what is actually implemented, tested, accepted, or released.

Status: MASTER TARGET DESIGN CANDIDATE - gameplay, art direction, sound design, and music direction consolidated; not a claim of current implementation or acceptance.

Purpose: This document defines the intended game clearly enough that design, engineering, art, animation, audio, UI, QA, and future collaborators can picture the same product and build toward it without substituting their own genre assumptions.

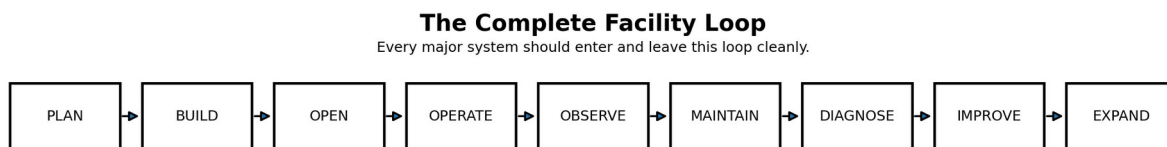
Authority rule: Live GitHub remains the authority for what is actually implemented, tested, accepted, or released. This GDD is the target product definition. If source and GDD differ, the difference is a product gap to resolve deliberately, not permission to pretend the target already exists.

One-sentence product definition: Self Storage Tycoon is a calm, touch-first construction-and-management simulation in which the player physically designs self-storage facilities, operates the businesses those designs create, maintains and improves them, and grows from hands-on owner-operator to multi-property storage company.

Core causal chain: LAYOUT -> OPERATIONS -> ECONOMICS -> GROWTH

PART I - THE GAME

1. The Game in One Mental Picture



Core game loop: the property turns layout into operations, economics, and growth.

Imagine the camera floating above a small self-storage property. The player can see the street, gate, office, drive-up doors, a covered loading area, cars, customers, carts, hallway entrances and the edges of an interior building. Time is paused.

The player taps **Build**, selects an interior building tool, and extends the building shell. They draw a main hallway, drag a row of 5x10 units along one side, flip the roll-up doors toward the hall, place a wide sliding loading door at the exterior wall, then put a cart corral beside that door. The game shows the cost and required connections before the player confirms.

The new wing takes time to construct. When the work finishes, the player sees that the hallway is dark and the climate zone has no capacity. They add lights and connect the area to HVAC. The wing becomes commissionable. The player opens it.

A customer arrives, enters a gate code, parks under the covered loading canopy, grabs a cart, pushes it through the wide door, waits briefly for an elevator, reaches Floor 2, follows signs down the hallway, opens the unit, unloads and leaves. Another customer arrives while the first cart is still upstairs. The corral runs low. The player can see the problem before opening any dashboard.

Later, the elevator's condition drops. Wait time grows. A Tech is already fixing a keypad, so the elevator waits in the maintenance queue. The player can hire another Tech, reprioritize work, add preventive maintenance, or accept slower service for a while. Upper-floor units become less attractive. The business impact can be traced back to a physical place.

That is Self Storage Tycoon.

The defining player promise

The player should be able to look at a problem and say:

****"I built this. I can see why it works or does not work. I can change it and watch the business respond."****

The property is not decorative scenery behind menus. It is the machine that customers, employees, equipment, money, and time flow through.

2. Design North Star and Non-Negotiables

The governing causal chain is:

LAYOUT -> OPERATIONS -> ECONOMICS -> GROWTH

The complete repeated loop is:

**PLAN -> BUILD -> OPEN -> OPERATE -> OBSERVE -> MAINTAIN -> DIAGNOSE -> IMPROVE -> EXPAND ->
REPEAT**

The game succeeds when a player can point to the property and explain a business result:

- "Those upper-floor units are harder to rent because this loading/elevator route is annoying."
- "I keep losing time because the carts end up on Floor 3."
- "The covered dock and second elevator made this wing much easier to use."
- "I built too much climate-controlled capacity for this market, so I am paying for HVAC I do not need."
- "My Porter spends half the day crossing the site because the maintenance room is in the wrong place."

The game fails if the facility is only a picture behind financial menus.

1. **No time pressure.** Pause, 1x, 2x, 4x. The player may stop indefinitely to think and build.
2. **Construction is real gameplay.** Units, buildings, hallways, loading, elevators, utilities, access, amenities and support spaces are placed physically.
3. **The property is the primary interface.** Important problems should first appear on the lot whenever the lot is the cause.
4. **Management is not detached from construction.** The player operates the machine they built.
5. **The tutorial is a functioning property.** It teaches through real needs instead of a disconnected checklist.
6. **Interior storage is central.** Drive-up remains important, but interior and multi-story logistics create much of the game's identity.
7. **Carts are a first-class logistics resource.** Cart location, supply and recovery matter.
8. **Elevators are a first-class capacity system.** Placement, capacity, queueing, condition and service matter.
9. **Security/access is physical.** Gates, keypads, controlled doors, cameras and lighting exist on the property.
10. **Utilities are readable abstractions.** They create design decisions without becoming engineering software.
11. **Problems are recoverable.** A mistake should become a diagnosis-and-improvement story, not a hidden death spiral.
12. **Scale changes the player's job.** Early play is hands-on; later play is delegation and capital allocation.

Calm but consequential

The game is deliberately non-twitch. The player always has **Pause, 1x, 2x, and 4x**. A difficult facility can create serious business problems, but it should never create artificial urgency that prevents the player from thinking.

The intended emotional rhythm is:

notice -> understand -> decide -> improve -> feel the property work better

not:

alarm -> panic -> click faster

3. Game Modes

Self Storage Tycoon has four complementary ways to play. They share the same simulation; they differ in starting state, goals, guidance, and freedom.

3.1 Guided Tutorial / First Facility

The tutorial is a real property, not a detached training room. The player inherits a partially established beginner facility and learns each mechanic only when the property creates a reason to care.

The tutorial ultimately becomes a normal owned property. It does not vanish when the lessons end.

3.2 Scenario Mode

Scenario Mode is a curated set of authored self-storage situations. Each scenario starts with a distinct property, market, constraint, or operational problem and asks the player to solve it using the real game systems.

Scenarios are not merely "easy / medium / hard" versions of the same map. Each should emphasize a different design problem.

Recommended initial scenario set:

Scenario	Starting situation	Main design question	Example completion condition
Maple Street Storage	Small partially developed beginner facility	Can you learn the complete owner-operator loop?	Stabilize and graduate the tutorial property
Blank Slate	Empty suburban parcel	Can you design a viable facility from nothing?	Open, lease, and reach sustainable operation
Downtown Vertical	Small expensive urban parcel	Can vertical density beat land scarcity?	Operate a successful multi-story property
Turnaround Property	Poor inherited layout, deferred maintenance	Can you diagnose and improve someone else's mistakes?	Restore service quality and profitability
The Full House	High occupancy, no spare capacity	How do you expand without breaking operations?	Add capacity while maintaining service
Mixed Storage	Drive-up + indoor + vehicle/RV demand	How do you allocate land and circulation?	Serve multiple product types successfully
Aging Facility	Profitable but obsolete infrastructure	What should be modernized first?	Complete a viable modernization plan
Premium Storage	Higher-end market	Can convenience and quality justify their cost?	Build and operate a premium facility sustainably

Scenario goals should use real state rather than arbitrary score-chasing. Valid goals include occupancy, sustained positive operating contribution, service quality, resolving a backlog, commissioning an expansion, reducing congestion, restoring infrastructure, or meeting a property-specific operational standard.

No normal scenario requires a countdown. The challenge is the situation, not the clock.

3.3 Sandbox Mode

Sandbox is the long-form construction-and-management toy.

Two core starts:

Empty Lot

Choose a parcel and market, receive starting capital, and build everything from the first entrance onward.

Existing Facility

Start with a generated or authored operating property and remodel, modernize, expand, or reposition it.

Standard Sandbox keeps:

- real construction cost;
- operating expenses;
- leasing and demand;
- staffing;
- maintenance;
- customer movement;
- logistics;
- save/resume;
- company progression where enabled.

Creative Sandbox can optionally provide:

- unlimited money;
- unlocked build catalog;
- instant or accelerated construction;
- optional incidents;
- optional maintenance;
- full simulation of routes, carts, elevators, customers, and staff.

Creative mode should still let the property *function*. It is not merely a scenery editor.

3.4 Company / Campaign Progression

The long-term progression connects properties into a storage company. The player can develop raw land, acquire existing facilities, appoint Managers, compare properties, allocate capital, and decide where the company grows next.

The campaign should never erase the detailed property game. It changes the player's job from doing routine work personally to designing systems and managing exceptions.

PART II - STARTING A PROPERTY

4. New Game: The Empty-Lot Experience

The default non-tutorial fantasy is starting with land and turning it into a functioning storage business.

4.1 New facility flow

CHOOSE MARKET -> CHOOSE PARCEL -> REVIEW CAPITAL -> ENTER PAUSED LOT -> PLAN ACCESS -> BUILD MINIMUM FACILITY -> COMMISSION -> OPEN -> LEASE -> OPERATE

The player should not be dropped onto empty asphalt with no context. Before entering the lot, the game shows only the information needed to make a meaningful start:

- parcel size and shape;
- road frontage;
- purchase/development cost;
- starting cash after acquisition;
- broad local demand by storage product;
- broad price level;
- whether the parcel favors drive-up, indoor, vertical, or mixed development;
- any obvious physical constraint.

4.2 Parcel variety

Parcels create replayability through physical and market differences rather than random gimmicks.

Potential parcel attributes:

- lot area;
- shape;
- road frontage;
- allowable entrance zones;
- acquisition cost;
- expansion opportunity;
- surrounding density;
- local demand mix;
- price level;
- climate-controlled demand share;
- vehicle/RV demand;
- utility capacity where useful;
- weather/climate profile where it materially changes design.

A large suburban rectangle should naturally support a different strategy from a narrow urban site.

4.3 Minimum viable facility

Before a property can open, the simulation verifies a small set of understandable requirements.

Target opening requirements:

- valid road/street connection;
- valid entrance/exit or combined access arrangement;
- customer route from entrance to at least one rentable product;
- at least one commissioned rent-ready unit;
- required office/customer-service capability for that property model;
- required access/security boundary;
- required utility service for any asset that depends on it;

- no route-blocking construction state that makes normal access impossible.

The UI should never simply say "Cannot Open."

Example:

****NOT READY TO OPEN****

- *No commissioned rent-ready units*
- *Office has no customer route*
- *Main gate is not connected to the street*

4.4 Office requirement and self-service progression

The beginner and traditional facility should require a physical office/reception function.

Later properties may unlock a reduced-office or self-service model using online rental, remote access management, and more automation. That should be a genuine operating strategy with tradeoffs, not a way to delete customer service from the game.

4.5 Recovery from a bad first build

A player should be able to make a poor first plan without soft-locking the save.

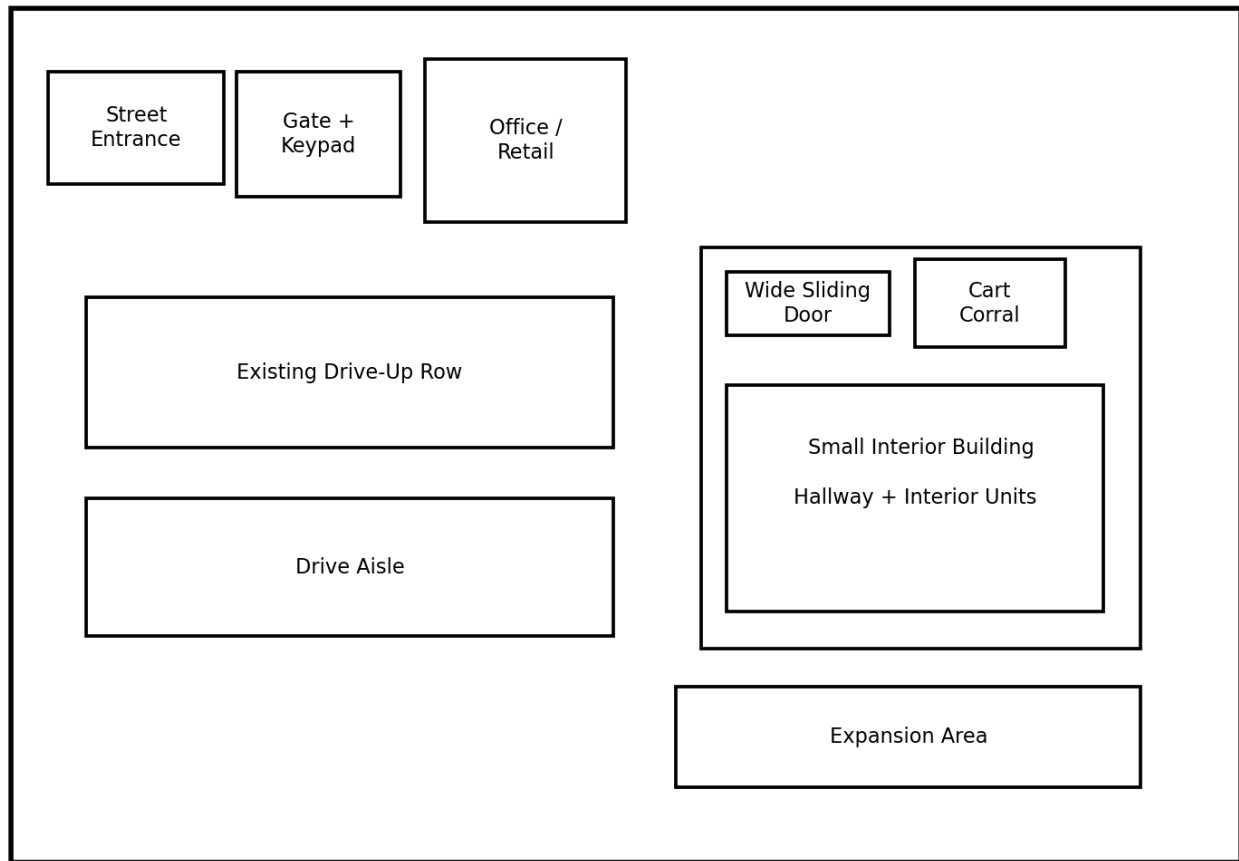
Protective mechanisms:

- placement preview before commitment;
- clear dependency warnings;
- bounded undo for accidental placements where safe;
- cancellation of unfinished construction with visible sunk cost;
- ability to close/renovate/demolish unoccupied assets;
- optional scenario/tutorial safety reserve;
- no hidden opening requirement discovered only after the player spends all cash.

PART III - THE TUTORIAL AS A REAL PROPERTY

5. Maple Street Storage

Tutorial Property: Maple Street Storage



Partially established, healthy-but-capped starter. Enough works to teach by improving it.

Maple Street tutorial-property schematic.

Maple Street is the player's first facility and tutorial. It is intentionally **partially established**, not empty and not broken beyond recognition.

5.1 Starting fiction

The player has taken control of a small storage property that is operational but capped. The previous owner maintained the basics, but the site is underdeveloped. There is a small office, a gate, a short drive aisle, one drive-up row, and a modest interior building. Most units are occupied. One recently vacated unit needs make-ready. There is expansion land along the rear/side edge.

The property begins **paused**.

5.2 First camera shot

The opening shot should allow the player to see, without panning:

- street edge;
- entrance/gate;
- office;

- drive-up row;
- loading entrance to the interior building;
- at least one customer/vehicle or recognizable activity marker;
- expansion space.

The player should immediately understand: "This is a place I own."

5.3 Authored first message

Welcome to Maple Street Storage.

The property is already open. Most units are rented. One unit needs to be turned over before it can rent again.

Take a look around. Time is paused until you're ready.

No countdown. No penalty for exploring.

5.4 Tutorial progression at Maple Street

The tutorial sequence is a chain of real property needs:

Beat	World creates the need	Mechanic taught	Visible proof
1	One unit is not rent-ready	inspect + Owner make-ready	Owner works; unit changes state
2	Rent-ready vacancy exists	leasing + asking rent	customer/lease appears
3	Daily costs continue between billing days	money comprehension	cash/ledger explain timing
4	Property reaches practical capacity	construction	new units appear physically
5	Interior customer uses loading entrance	indoor access	route is visible
6	Customer needs a cart	carts/corral	cart leaves and returns
7	A service object needs work	maintenance	task and repair are visible
8	Owner becomes capacity constraint	first employee	delegated work completes
9	Weak area exists	camera/light/keypad upgrade	coverage/access improves
10	Climate demand is worth serving	HVAC/climate	new product becomes viable
11	Land pressure motivates vertical design	elevator/floor concept later	multi-story property unlock

The tutorial ends when the player has proven the complete build-operate-improve loop, not when a fixed number of popups have been dismissed.

Maple Street's job in the game

Maple Street is not a perfect facility. It is a readable facility with deliberate opportunities.

The property should contain examples of:

- something already working;
- something waiting for the Owner;
- an obvious expansion space;
- a small interior-storage opportunity;
- basic but incomplete security;
- a small number of carts;
- enough customers to make the lot feel alive;
- enough cash to teach decisions without creating a bankruptcy trap.

6. The First 45-60 Minutes

This section is intentionally authored at moment-to-moment level so the development team can picture the game.

6.1 Minute 0-5: Learn the property

The game opens paused at Maple Street. A gentle focus pulse highlights the unit with pending make-ready only after the player pans or taps once.

The player taps the unit. Its inspection card says:

Unit 107 - 5x10

Vacant - Make-ready required

Last tenant moved out yesterday

Work: sweep, inspect door, reset unit

Assigned: nobody

Buttons:

- **Start Owner Make-Ready**
- Close

The player starts the task. The Owner appears or a clear Owner-work representation begins at that unit. The player is invited to press 1x, 2x or 4x. When the task completes, the door/state presentation changes and the unit becomes Rent Ready.

6.2 Minute 5-10: First business result

The game explains only what matters now:

***Rent-ready units can lease.** Asking rent affects how strongly the market responds. There is no single perfect price.*

A prospect/market lease opportunity occurs naturally. The player sees a customer/vehicle arrive or a readable lease event tied to the property.

6.3 Minute 10-15: Money without a spreadsheet lecture

The business view shows:

- cash;
- monthly rent roll;
- daily operating cost;
- recent collected rent;
- construction spending.

Copy explicitly distinguishes monthly billing from daily expenses.

The tutorial does not require the player to make a profit within an arbitrary seven-day window.

6.4 Minute 15-25: First expansion

The facility becomes full or near full while market demand remains. The property itself looks capped: no rent-ready units, customers still inquire, and the Growth view points to physical expansion options.

The player selects **Build** -> **Drive-Up Storage** -> 5x5 or another offered starter product. The player extends the existing drive aisle and drags a short row of units. The preview displays:

- number of units;
- cost;
- aisle/access validity;
- expected ongoing operating cost;
- construction/commission requirement.

The player confirms and watches construction complete. Then they commission it.

6.5 Minute 25-35: Interior logistics

The player is asked to inspect the existing interior building rather than build a new complex from scratch.

A customer arrives at the loading area. The player sees the sequence:

vehicle -> loading area -> cart corral -> wide door -> hallway -> unit.

If a cart is available, the trip is smooth. If not, the customer visibly waits or searches according to the simplified customer behavior model.

The player taps the cart corral and sees:

Main Loading Cart Corral

3 carts assigned

2 available

1 in use

Target stock: 2+

No abstract "cart satisfaction" number is required for basic diagnosis.

6.6 Minute 35-45: First maintenance problem

A light, keypad, roll-up door or other simple asset develops a service need. The problem appears where it exists. The player can tap it and start Owner maintenance.

If the Owner is already busy, the task queues. This teaches service backlog naturally.

6.7 Minute 45-60: Delegation

The player's facility is now busy enough that doing every task personally creates visible delay. The game introduces the first hire.

The hire lesson is not "Click Hire because the tutorial says so." It is:

Two jobs are waiting and the Owner can only work one at a time. Hiring help adds service capacity.

The player hires a basic employee/Porter. One task is delegated. The player watches two pieces of work progress in parallel.

The first session ends with the player understanding:

I built something -> customers used it -> work appeared -> I handled or delegated it -> the business changed.

7. Tutorial Chapters Beyond the First Hour

The tutorial is not over when the player knows the buttons. It gradually becomes the campaign.

65.1 Chapter 1 - Take Control

Teach:

- inspect;
- time;
- make-ready;
- rent;
- money timing.

65.2 Chapter 2 - Add Capacity

Teach:

- build preview;
- drive aisle;
- row construction;
- commissioning;
- construction cash impact.

65.3 Chapter 3 - Make Interior Storage Work

Teach:

- loading;
- wide door;
- cart corral;
- carts;
- hallway access;
- customer journey.

65.4 Chapter 4 - Keep the Property Working

Teach:

- maintenance;
- service backlog;
- cleaning;
- first employee;
- task delegation.

65.5 Chapter 5 - Improve Quality

Teach:

- lighting;
- cameras;
- keypad/access;
- customer complaint/thought;
- targeted upgrade.

65.6 Chapter 6 - Add Climate

Teach:

- HVAC capacity;
- coverage;
- power cost;
- climate product;
- commissioning.

65.7 Chapter 7 - Build Up

Teach:

- second floor;
- stair;
- elevator shaft;
- elevator capacity;
- upstairs carts;
- vertical congestion.

65.8 Chapter 8 - Become the Manager

Teach:

- area assignment;
- priorities;
- preventive maintenance;
- operating review;
- less direct Owner work.

65.9 Graduation

Maple Street tutorial/campaign graduates when the player has:

- built;
- leased;
- maintained;
- delegated;
- diagnosed;
- recovered;
- expanded;
- operated a stable property through multiple billing/turnover cycles.

No timer is required.

The tutorial uses deterministic mastery milestones, not subjective "play well enough" gates.

Relevant mastery milestones include:

- first unit restored to rent-ready;
- first lease after turnover;
- first expansion commissioned;
- first successful cart-supported interior trip;
- first delegated task completed;
- first diagnosed and repaired service fault;
- first viable climate area commissioned;
- first functioning upper floor opened.

The player can remain on Maple Street after graduation.

PART IV - BUILDING THE FACILITY

8. Construction Grammar

This is the most important clarification in v0.2. The player does not merely buy capacity. The player composes a facility from a consistent set of physical primitives.

4.1 Construction hierarchy

PARCEL

- > exterior circulation
- > building shell
- > floor
- > hallway / room geometry
- > storage units
- > doors and vertical connectors
- > customer/service infrastructure
- > utilities and security
- > furniture/equipment/amenities
- > commissioning

Each layer creates requirements for the layer above it.

4.2 Touch interaction grammar

The default mobile interaction is:

CATEGORY -> TOOL -> PLACE/DRAW -> LIVE PREVIEW -> CONFIRM

Common gestures:

- **Tap** selects an object or starts a single placement.
- **Drag** draws linear geometry such as an aisle, wall, hallway or row of units.
- **Drag handles** resize a rectangular room/building where allowed.
- **Rotate** changes orientation.
- **Flip door side** changes which frontage a unit opens toward.
- **Duplicate** repeats a selected unit/row/room pattern.
- **Paint** assigns a zone or coverage designation where a zone tool is appropriate.
- **Long press** opens precision/context actions rather than hiding them behind a desktop-only right click.

Every build preview must answer four questions before confirmation:

- Is it valid?
- What will it cost?
- What does it require to function?
- What will it connect to?

4.3 Construction states

PLANNED -> COMMITTED -> UNDER CONSTRUCTION -> BUILT/UNCONNECTED -> READY TO COMMISSION -> OPERATING

The player should visually distinguish all six states.

4.4 Building shell construction

A building shell is drawn as a footprint. Later properties may support irregular shapes, but the first implementation can use rectangles and additive extensions.

The player sets:

- footprint;
- number of floors available/authorized;
- exterior doors;
- loading interfaces;
- interior floor plan.

A shell alone earns no rent.

4.5 Hallway construction

Hallways are not decorative empty tiles. They create the pedestrian/cart movement network inside a building.

Recommended gameplay widths:

- **Narrow:** cheapest; poor for opposing carts; suitable only for low traffic.
- **Standard:** default; comfortable passing and ordinary unit rows.
- **Wide:** expensive space sacrifice; useful near elevators/loading/intersections.

The exact dimensions remain tuning variables. The player-facing point is the tradeoff between rentable area and circulation quality.

4.6 Unit-row tool

The most important build convenience is a **row placement tool**.

Example flow:

1. Player chooses **5x10 Interior Unit**.
2. Player drags along a hallway-facing wall.
3. Ghost units repeat automatically at valid spacing.
4. Doors face the hallway by default.
5. Player can flip door orientation before confirming.
6. Live preview displays count, total construction cost, expected product type, and any blocked doors.
7. Confirm creates the committed construction order.

The player can still place single units for irregular spaces.

4.7 Copy and repeat

The game should support copying a proven pattern once the player has built it manually at least once. Repeating a 12-unit row should not require 12 separate taps.

4.8 Why this grammar matters

This construction grammar is what makes the title an architect/tycoon game rather than a capacity-management interface. The same grammar must work for a tiny tutorial building and a mature multi-story property.

Canonical construction hierarchy

PARCEL

- > exterior circulation
- > building shell
- > floor

- > interior circulation
- > storage/product spaces
- > loading and vertical circulation
- > utilities/security
- > amenities/service spaces
- > commissioning
- > operation

This hierarchy matters because later systems should not invent incompatible spatial meanings.

9. Complete Buildable-Object Catalog

The build palette should ultimately be able to represent a believable self-storage property without asking the player to construct meaningless detail.

52.1 Site and perimeter

Buildable or placeable:

- street-connected entrance;
- exit;
- combined entrance/exit;
- perimeter fence;
- pedestrian gate where useful;
- vehicle gate;
- gate pedestal/keypad;
- monument sign;
- directional sign;
- asphalt/paved circulation;
- parking spaces;
- loading spaces;
- turnaround/apron;
- landscaping strips or planters where they affect presentation/quality.

Simulation purpose:

- defines where vehicles and customers may enter;
- controls traffic topology;
- establishes security boundary;
- consumes parcel area;
- can create or solve congestion.

Not simulated:

- civil-engineering drainage calculations;
- curb radii by code book;
- survey-grade grading;
- individual pavement layers.

52.2 Building shell and structural layer

Buildable:

- ground-floor shell;
- additive shell extension;
- upper floor;
- exterior wall;

- interior partition wall;
- roof implied by enclosed shell;
- exterior service/loading opening;
- stair;
- elevator shaft.

The shell defines enclosed space. Enclosed space is not automatically rentable.

A valid building must have:

- connected floor geometry;
- at least one legal access route;
- no impossible overlap with exterior roads or another structure;
- vertical support/shaft continuity for upper floors;
- sufficient exit/access abstraction according to the game's simplified rules.

The game should explain invalidity in plain language:

- "This door opens into a wall."
- "Floor 2 has no stair or elevator connection."
- "These units have no hallway frontage."
- "This extension blocks the only vehicle route."

52.3 Interior circulation

Buildable:

- narrow hallway;
- standard hallway;
- wide hallway;
- elevator lobby;
- loading vestibule;
- interior doorway;
- wide sliding door;
- automatic sliding door;
- access-controlled door.

The hall network is the primary pedestrian/cart graph.

52.4 Storage products

Core:

- 5x5;
- 5x10;
- 10x10;
- 10x20.

Access variants:

- drive-up;
- interior ground floor;
- interior upper floor.

Environment variants:

- standard;
- climate-controlled.

Future products should only be added when they create a different spatial or business decision.

52.5 Loading and customer logistics

Buildable:

- uncovered loading bay;
- covered loading canopy;
- covered loading dock;
- cart corral;
- waiting/bench area where justified;
- wayfinding sign;
- automatic/wide loading door.

52.6 Office and customer facilities

Buildable:

- office/reception;
- retail/lock-and-box area;
- restroom;
- water fountain;
- small waiting area;
- staff-only office space later.

52.7 Service and staff facilities

Buildable:

- janitorial closet;
- maintenance/workshop room;
- staff break room;
- dumpster/cleanout court;
- equipment/service room;
- security/control room on advanced sites.

52.8 Security and access

Buildable:

- gate keypad;
- door keypad;
- camera;
- exterior light;
- interior light;
- higher-grade gate/fence upgrades;
- later controlled wing/floor boundary.

52.9 Utilities

Buildable or upgradeable:

- electrical service/capacity upgrade;
- electrical/service room where presentation benefits;
- HVAC plant;
- additional HVAC capacity;
- water service for building amenities.

The player places major equipment and assigns/reads coverage. The player does not draw wires, pipes, refrigerant lines, or ducts one segment at a time.

Parking and staging

Parking is mechanically light but real.

Parking/loading destinations can include:

- office/customer parking;
- ordinary tenant parking;
- loading stalls;
- covered loading bays;
- staff/service parking as an abstraction where needed.

The game does not simulate parking-lot etiquette in detail. Parking exists primarily to:

- provide valid destinations;
- consume land;
- interact with drive aisles;
- create or resolve circulation conflict;
- distinguish office visits from loading activity.

A car should not stop in the middle of a drive aisle simply because no destination system exists.

10. Placement, Validation, Batch Building, and Feedback

Construction must never feel like guessing.

53.1 Before confirmation

Every ghost placement uses four simultaneous signals:

- **Geometry:** footprint and orientation.
- **Validity:** green/valid, red/invalid, with a specific reason.
- **Dependencies:** access, hallway frontage, power, HVAC, security, loading, or vertical connection requirements.
- **Economics:** immediate build cost, estimated construction duration, and ongoing operating cost when meaningful.

53.2 Validation classes

A placement can be:

VALID NOW

Can be committed and will be capable of commissioning when construction completes.

VALID BUT INCOMPLETE

May be built, but cannot operate until another requirement is supplied. Example: a climate wing can be constructed before enough HVAC capacity exists.

INVALID

Cannot be committed because geometry or authoritative rules make it impossible.

This distinction prevents the game from blocking useful staged construction while still telling the truth.

53.3 Dependency examples

A 5x10 interior unit can be physically built if:

- it lies inside valid building space;
- its door faces valid hallway frontage;
- it does not overlap another object.

It becomes commercially commissionable only if:

- the hallway connects to an accessible building entrance;
- vertical access exists if it is on an upper floor;
- required power/lighting abstractions are satisfied if the design requires them;
- HVAC coverage exists if the unit is climate-controlled.

53.4 Batch placement

Row tools must preview:

- count;
- total cost;
- total rentable area;
- blocked or invalid members;
- door orientation;
- product/environment type.

The player may confirm valid members only if the behavior is explicit; default should be all-or-nothing for a row so accidental partial builds do not surprise the player.

Player-facing placement contract

Before the player commits money, the preview must answer:

- what will be built;
- where doors/frontage face;
- whether it is valid;
- what requirement is missing if incomplete;
- how many objects will be created;
- immediate cost;
- construction duration where relevant;
- recurring operating cost where relevant.

Construction should feel like planning, not trial-and-error error messages.

11. Edit, Cancel, Undo, Renovate, Convert, and Demolish

A construction game needs a clear answer for mistakes.

54.1 Planned objects

Before commitment:

- freely move;
- rotate;
- resize where allowed;
- cancel with no cost.

54.2 Under-construction objects

The player may cancel construction subject to a visible sunk-cost rule.

The exact refund rate is a tuning question, but the preview must state it before confirmation.

54.3 Recently committed placement

A short bounded Undo can exist for accidental taps if technically safe. Undo is not a loophole for speculative free construction.

54.4 Operating assets

Possible actions:

- upgrade in place;
- convert product/environment where supported;
- close temporarily for renovation;
- demolish;
- relocate only for assets that are realistically movable, such as a cart corral, sign, camera, or some furniture.

Buildings and occupied units should not teleport.

54.5 Occupancy protection

The game must not let the player demolish or materially convert an occupied unit without first resolving its lease through a supported gameplay process.

54.6 Demolition consequence

Demolition can involve:

- direct cost;
- downtime;
- lost capacity;
- salvage/refund if design allows;
- temporary route disruption.

The player sees the consequence before confirming.

12. Construction Time and Contractors

Construction should exist in time but should not become contractor micromanagement.

55.1 Construction orders

A committed project creates a construction order with:

- scope;
- cost;
- start day;
- expected duration;
- current phase;
- route/closure impact if any;
- commissioning requirements.

55.2 Construction phases

A simple universal model is preferred:

SITE / SHELL -> INTERIOR / EQUIPMENT -> COMPLETE -> COMMISSION

Small objects can collapse this to a short install.

55.3 Who builds?

The default assumption is contracted construction labor. The player purchases and schedules the work; they do not individually hire carpenters, electricians, or plumbers.

Construction crews can appear visually without becoming permanent staff entities unless later play proves contractor management is fun.

55.4 Operating around construction

Construction may:

- close a hallway;

- occupy a loading bay;
- restrict a drive aisle;
- create temporary service inconvenience.

The game must preview closures before the player commits.

55.5 No real-time waiting trap

At 4x, ordinary first-facility construction should complete fast enough to remain part of a session. Large buildings can take longer, but the player should always have meaningful operating decisions while they build.

Construction crews are visualized where useful, but permanent contractor micromanagement is out of scope unless later playtesting proves it fun.

PART V - CAMERA, FLOORS, AND INTERIORS

13. Camera, Floors, Roofs, and Cutaway

Multi-story construction is impossible to understand without a defined camera model.

56.1 Camera

Touch controls:

- one-finger drag: pan;
- pinch: zoom;
- optional two-finger rotate if isometric rotation is supported;
- tap: select;
- long-press: precision/context;
- double-tap selected object: focus camera.

The game should not require free 3D camera dexterity.

56.2 Floor selector

When a multi-story building is selected, the player sees a compact floor selector:

- Ground / 1;
- 2;
- 3;
- etc.;
- All / Exterior view where useful.

56.3 Cutaway

When inspecting an interior floor:

- roof hides automatically;
- floors above become hidden or translucent;
- exterior walls may cut away where they block the selected floor;
- the currently active floor remains visually dominant.

56.4 Vertical alignment

When placing:

- elevators;
- stairs;
- shafts;
- major vertical service elements,

the floors above/below show alignment ghosts.

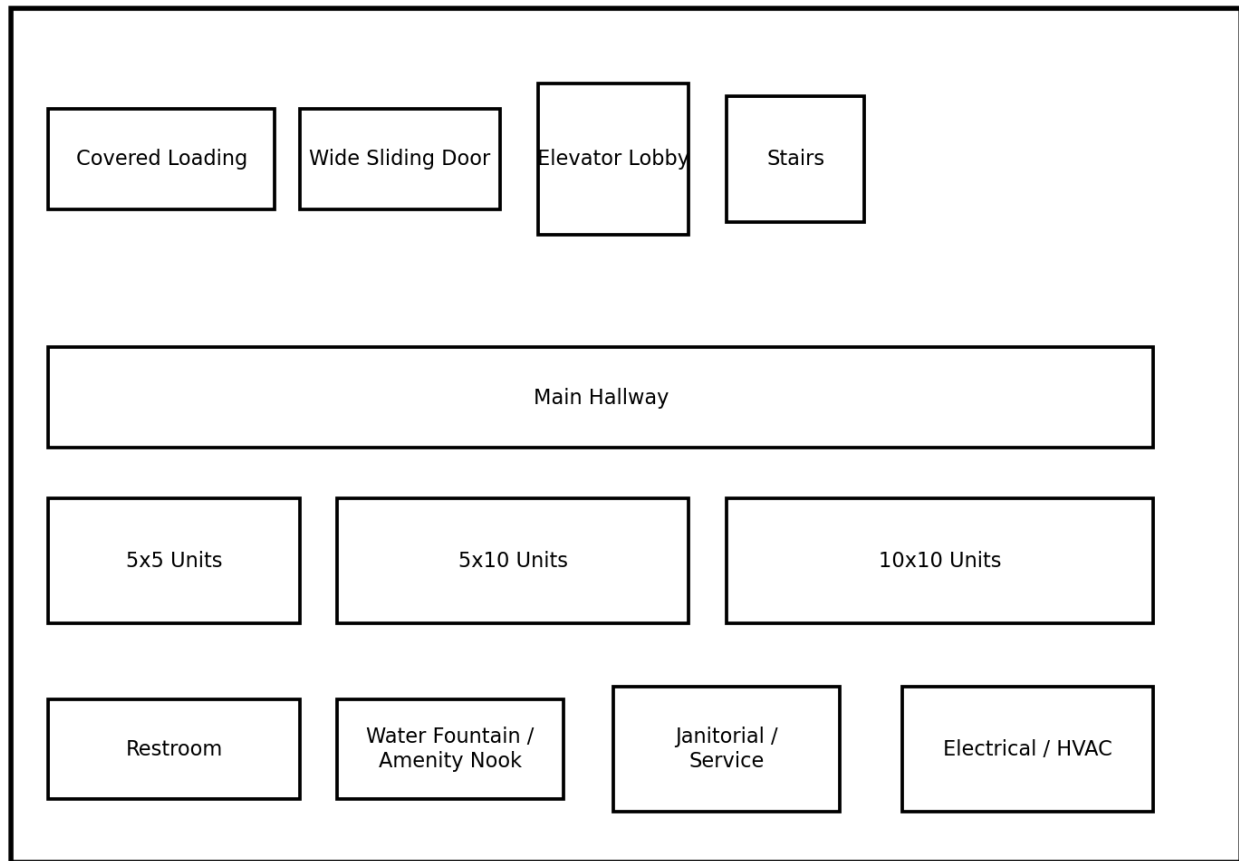
An elevator shaft cannot silently shift sideways between floors.

56.5 Exterior-to-interior transition

Tapping a building or selecting an interior customer can smoothly enter the cutaway view. Exiting returns to the property overview without losing orientation.

14. Interior Building Anatomy

Interior Building - Example Floor



Example only: the game should make the loading -> cart -> elevator -> hallway -> unit chain legible.

Example interior-floor anatomy and circulation relationships.

An interior storage building is a machine composed of connected systems.

7.1 Primary chain

LOADING AREA -> BUILDING DOOR -> CART -> ELEVATOR/STAIRS -> HALLWAY -> UNIT

Every link can be inspected separately, but the player should be able to see the whole journey.

7.2 Floor plan decisions

The player chooses:

- main hallway path;
- hallway width;
- unit sizes and door orientation;
- elevator/stair placement;
- service-room placement;
- restroom/water-fountain placement where used;
- access-controlled boundaries;
- lighting/camera placement;
- climate zone coverage.

7.3 Rentable efficiency vs usability

Packing the floor with units maximizes rentable area but can create poor circulation, long routes, weak service access or elevator congestion. Wide halls and generous lobbies improve movement but sacrifice rentable square footage.

That tradeoff is a core source of architectural gameplay.

Interior spatial grammar

A typical interior customer journey is:

LOADING / ENTRY -> CART -> CONTROLLED DOOR -> ELEVATOR OR HALL -> MAIN HALL -> SECONDARY HALL -> UNIT

A well-designed floor should make that route short, legible, and robust.

A poor design can still function, but creates visible consequences:

- long walks;
- cart congestion;
- elevator demand;
- confusing intersections;
- service distance;
- poor loading relationships.

15. Hallways, Doors, Routing, and Wayfinding

The interior building needs consistent traffic semantics.

57.1 Movement networks

Target networks:

- vehicle;
- pedestrian;
- cart-accessible pedestrian;
- staff/service;
- vertical connector.

They may share graph edges, but capability is explicit.

57.2 Hallway width semantics

Narrow

- cheapest and space-efficient;
- one cart may slow opposing movement;
- poor near heavy loading/elevator demand.

Standard

- normal default;
- ordinary passing;
- supports most unit rows.

Wide

- higher space cost;
- improved flow near loading, elevators, intersections, or premium lobbies.

The game does not need collision physics for every shoulder. It needs route capacity/friction that produces understandable congestion.

57.3 Door throughput

Doors contribute service time/capacity:

- standard door;
- wide manual sliding door;
- automatic wide door.

A building with hundreds of units behind one narrow door should behave differently from one with a proper loading entrance.

57.4 Intersections

Busy intersections can accumulate friction. The player should not need traffic lights inside a storage building; route design and width are enough.

57.5 Wayfinding

For simple floors, shortest valid route is enough.

For complex buildings, wayfinding signs can reduce wandering or navigation penalty. The game should keep this effect modest and explainable.

A hallway is not simply "walkable floor." It is a capacity-bearing part of the building.

The simulation should not model shoulder-to-shoulder collision physics. It should model enough friction and capacity that narrow or badly concentrated circulation behaves differently from a good layout.

PART VI - STORAGE PRODUCTS

16. Storage Product Model

18.1 Drive-up units

Strengths:

- excellent direct access;
- simple customer journey;
- easy to understand;
- strong for large/frequent-access customers.

Costs/tradeoffs:

- consumes land and aisle frontage;
- requires vehicle circulation;
- lower density than multi-story interior.

18.2 Interior standard units

Strengths:

- efficient land use;
- dense unit count;
- weather protection.

Dependencies:

- loading;
- door access;
- hallways;
- carts;
- lighting/security;
- elevators if upper floor.

18.3 Climate-controlled units

Additional dependencies:

- HVAC capacity;
- power;
- enclosed coverage zone;
- climate equipment condition.

18.4 Unit sizes

Core initial sizes:

- 5x5;
- 5x10;
- 10x10;
- 10x20.

Additional sizes should be introduced only when they create new market/layout decisions.

Product dimensions

Every rentable product is defined by a small set of meaningful attributes:

- size;

- drive-up vs interior;
- floor;
- standard vs climate-controlled;
- access convenience;
- security/access class where applicable;
- current condition;
- current readiness;
- asking rent;
- lease state.

17. Unit Design Contract

Every storage unit instance or unit type must expose the following conceptual contract.

Field	Meaning
Product size	5x5, 5x10, 10x10, 10x20, future sizes
Access type	drive-up / interior / upper-floor
Door orientation	where customer/staff enter
Route node	authoritative target for access/pathing
Environment	standard / climate
Commission state	whether it may enter inventory
Commercial state	vacant, rent-ready, occupied, service-restricted, etc.
Condition	wear/service state where authoritative
Lease	current contract if occupied
Asking rent group	product pricing group
Loading burden	derived from route/loading/elevator chain
Security/lighting context	derived coverage where used

This is the shared language connecting construction, operations and business.

A unit's quality should arise from the systems around it rather than from a hidden arbitrary star rating.

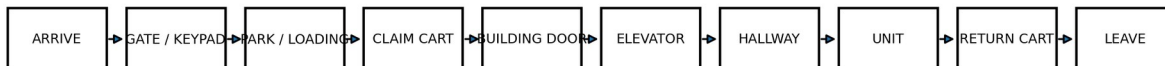
Derived influences can include:

- distance to loading/access;
- floor/elevator dependency;
- climate availability;
- security coverage;
- lighting;
- cart convenience;
- hallway route;
- physical condition.

PART VII - CUSTOMER LOGISTICS

18. Customer Move-In Journey

Interior Customer Move-In Journey



Physical move-in journey from arrival to unit.

The move-in journey is one of the best ways to make the facility feel alive.

8.1 Arrival

The customer vehicle enters through a valid gate route. If the facility has access control, the vehicle pauses at the keypad/gate.

8.2 Gate / keypad

The customer enters their access credential. The gate opens after the access-control service time. A slow or failing keypad can create visible delay.

8.3 Parking / loading

Drive-up customer: proceeds directly toward the unit if valid.

Interior customer: seeks an appropriate loading/parking location.

8.4 Cart claim

If the visit requires a cart, the customer goes to the nearest appropriate corral with an available cart. If no cart is available, the customer can:

- wait briefly for a return;
- walk to another nearby corral;
- proceed without one at a slower service rate if plausible.

The exact behavior is tuning, but cart scarcity must be visible and consequential.

8.5 Building access

The customer uses a wide sliding/automatic door or standard controlled-access door. A keypad may be required at the building boundary.

8.6 Elevator

If the unit is upstairs, the customer joins the elevator queue. Carts consume elevator capacity.

8.7 Hallway and unit

The customer follows the hallway path to the unit. Wayfinding quality can influence wandering on complex floors later.

8.8 Return path

The customer leaves the unit, returns toward loading, and ideally returns the cart. Not every cart is returned perfectly; this creates cart-recovery work.

Move-ins and move-outs are the most demanding ordinary customer trips and therefore expose weak facility design.

19. Loading Infrastructure

11.1 Basic loading area

A marked exterior space where an interior-storage customer can stop and unload.

11.2 Covered loading area

Provides weather protection and a more legible transition into the building. It can support stronger customer convenience and premium property quality.

11.3 Covered loading dock

A larger built structure that can serve multiple vehicles and concentrate carts, wide doors and elevator access.

It has real capacity:

- vehicle bays;
- approach route;
- cart staging;
- building-door throughput.

11.4 Wide sliding / automatic doors

Door types influence:

- opening time;
- cart/pedestrian throughput;
- maintenance workload;
- access-control integration;
- power requirement for automatic doors.

The player's visual read should be immediate: a premium loading entrance looks and operates differently from a single narrow door.

Loading hierarchy

Possible loading progression:

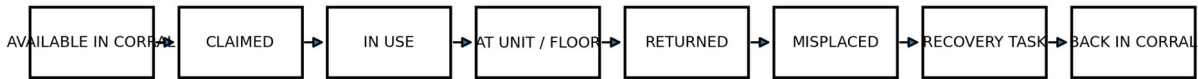
1. basic exterior loading space;
2. marked loading area;
3. covered loading canopy;
4. covered loading dock;
5. multiple-bay loading area for large properties.

The loading system connects:

vehicle -> unloading -> cart -> building entrance -> elevator/hall -> unit

20. Cart System

Cart Lifecycle



Cart lifecycle and recovery loop.

58.1 Cart lifecycle

IN CORRAL -> CLAIMED -> IN USE -> RETURNED / MISPLACED -> RECOVERED -> IN CORRAL

Optional condition adds:

DAMAGED -> OUT OF SERVICE -> REPAIRED

58.2 Cart acquisition

The player buys carts as capital equipment.

A purchase specifies:

- quantity;
- unit cost;
- destination/home corral.

Delivery can be immediate or short-delay depending on pacing.

58.3 Customer cart need

Not every routine access trip must require a cart.

Visit profiles can include:

- no cart;
- cart preferred;
- cart required for efficient move.

Move-ins/move-outs should strongly favor carts.

58.4 Claim behavior

A customer searches within a bounded reasonable area:

1. preferred/nearest corral;
2. alternate nearby corral;
3. wait briefly;
4. continue without cart only if that visit can plausibly do so.

The inspection UI should expose the resulting delay, not a hidden "cart happiness" number.

58.5 Elevator interaction

A person with cart consumes more cabin capacity and loading/dwell time.

58.6 Return behavior

Return outcome is influenced by:

- nearby visible corral;
- route;
- customer behavior abstraction.

The game should avoid cartoonishly constant abandonment. Misplacement exists to create occasional logistics work.

58.7 Rebalancing

Early:

- Owner/Porter receives explicit recovery task.

Mid:

- player sets target stock per corral.

Late:

- Manager/Porter policies rebalance automatically within capacity.

58.8 Cart inspection card

Example:

Platform Cart #12

Status: Available

Location: Main Loading Corral

Home: Main Loading

Condition: Good

Last use: Day 214, 2:14 PM

Or pooled presentation:

Main Loading Carts

4 present / 6 assigned

1 in use

1 misplaced on Floor 2

Target stock: 4

58.9 Cart failure cases

- no carts;
- too many carts clogging a tiny corral;
- carts stranded upstairs;
- damaged cart;
- corral too far from loading/elevator.

Every failure has a physical tell.

Cart design rule

Carts are physical shared resources, not a passive amenity score.

A customer should be able to arrive at an empty corral because carts are:

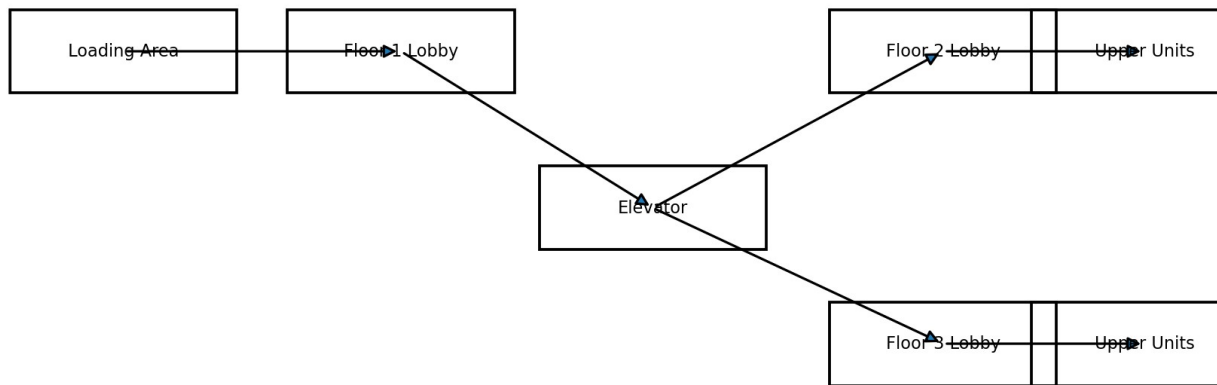
- in use;
- stranded elsewhere;
- damaged;
- assigned to another location.

The player should be able to fix the situation through:

- more carts;
- better corral placement;
- rebalancing;
- Porter capacity;
- facility redesign.

21. Elevator and Stair System

Elevator as a Capacity-Constrained Connector



Elevator capacity and vertical customer flow.

59.1 Elevator state

Possible states:

- Idle;
- Loading;
- Moving;
- Unloading;
- Queued demand;
- Maintenance warning;
- Out of service.

59.2 Dispatch abstraction

The player does not manually dispatch cars.

For one elevator:

- serve queued floor requests automatically.

For multiple elevators:

- system assigns the next appropriate car using a simple authoritative dispatch rule.

The player manages capacity and placement, not elevator buttons.

59.3 Queue inspection

At any lobby, show:

- people waiting;
- carts waiting;
- recent average wait;
- elevator status;
- capacity;
- service warning.

59.4 Peak behavior

Move-in/move-out visits consume more elevator capacity than routine access. This creates natural peaks without scripted disasters.

59.5 Outage

An outage:

- blocks cart-based movement to affected upper floors;
- may permit pedestrian stair access;
- creates a high-priority Tech/vendor task;
- reduces upper-floor convenience;
- can delay move-ins/outs.

The player sees affected floors immediately.

59.6 Redundancy

A second elevator:

- raises capital and maintenance cost;
- consumes floor area;
- improves throughput;
- provides redundancy.

This is a major architect decision, not an automatic upgrade.

Elevator design rule

The player does not manage elevator calls manually. The player manages:

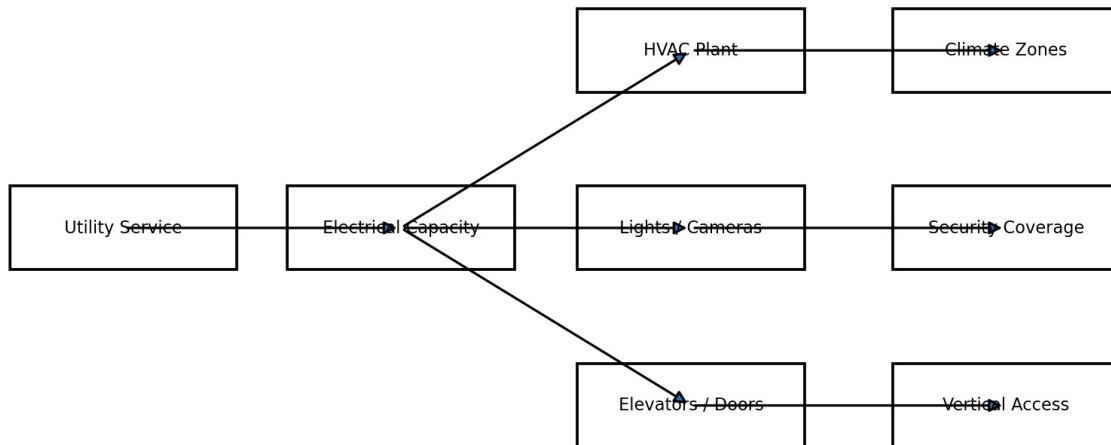
- how many elevators exist;
- where they are;
- what floors they serve;
- their capacity;
- their condition;
- whether the building has enough redundancy.

Stairs preserve pedestrian access and resilience but are not a substitute for cart/freight movement.

PART VIII - BUILDING SYSTEMS

22. Utilities and Infrastructure Abstraction

Readable Infrastructure Abstraction



Readable infrastructure abstraction: capacity and coverage rather than wire/pipe micromanagement.

The game should simulate **capacity and coverage**, not individual wiring, pipes or ducts.

13.1 Power

Model:

UTILITY SERVICE -> ELECTRICAL CAPACITY -> POWERED ASSETS

The player sees:

- total capacity;
- used capacity;
- which major assets depend on power;
- which area loses function if a service problem exists.

The player does not draw every wire.

13.2 HVAC

Model:

HVAC PLANT -> CAPACITY -> COVERAGE ZONE -> CLIMATE-CONTROLLED UNITS

The player places/assigns equipment and paints or inherits a coverage zone. The game shows whether the zone has enough capacity.

13.3 Security

Model:

- camera -> visible coverage cone/area;

- light -> illumination radius/zone;
- keypad -> access boundary;
- gate -> controlled vehicle boundary.

13.4 Water

Model:

BUILDING WATER SERVICE -> RESTROOM / FOUNTAIN AVAILABILITY

Water remains intentionally light unless play proves deeper plumbing would be fun.

Power

UTILITY SERVICE -> CAPACITY -> POWERED EQUIPMENT

Power affects gates, keypads, lights, cameras, elevators, HVAC, automatic doors, and office systems.

The player upgrades capacity and places major service equipment where useful. They do not draw individual wires.

HVAC

HVAC PLANT -> CAPACITY + COVERAGE -> CLIMATE-CONTROLLED SPACE

Climate-controlled units cannot simply be toggled from standard to climate without valid HVAC support.

Water

Water remains a light abstraction:

- building water service;
- restrooms;
- water fountains;
- future cleaning/service uses only if they become meaningful.

The player does not draw plumbing lines.

23. Security and Access Control

14.1 Gate keypad

Customers stop, enter a credential and wait for the gate to authorize/open.

Upgradeable dimensions:

- reliability;
- response/open time;
- audit/monitoring quality;
- weather resistance.

14.2 Building keypads

Interior buildings can require a second access point. This is especially valuable for climate-controlled or after-hours buildings.

14.3 Zone-specific access

Later properties may support controlled floors/wings. This should unlock only if it creates real security or operating decisions.

14.4 Camera coverage

Cameras have explicit coverage. The player can inspect blind spots. Cameras affect security quality/risk through authoritative simulation rather than random "crime chance" disconnected from the lot.

14.5 Lighting

Lighting has both customer-comfort and security value. Poorly lit exterior/interior routes should be visible at night.

Security should be meaningful without becoming a crime-management game.

62.1 Security inputs

Potential authoritative inputs:

- perimeter continuity;
- gate/access control;
- camera coverage;
- lighting coverage;
- operating/access hours;
- unresolved equipment failure;
- later staff/security presence.

62.2 Security output

The simulation may produce a security-quality/risk measure used by:

- customer perception;
- premium-product suitability;
- incident probability where justified.

62.3 Incident design

A security incident must:

- arise from understandable conditions;
- have a visible location;
- create a bounded consequence;
- produce a response task or decision;
- avoid random catastrophic theft detached from player agency.

62.4 Keypad failure

Failure types can include:

- degraded response;
- offline/faulted state.

The result is access delay or blocked controlled entry until repaired/bypassed by supported policy.

62.5 Cameras

A camera can be:

- operating;
- degraded;
- offline.

Coverage is geometric and inspectable.

Security systems should create readable differences without becoming a stealth/crime simulator.

24. Bathrooms, Water Fountains, and Small Amenities

These are real property objects, but they must remain lightweight.

15.1 Restrooms

Restrooms support:

- customer dignity/comfort;
- staff needs;
- facility-quality/inspection goals on larger sites.

They require cleaning and water service.

15.2 Water fountains

A water fountain:

- requires building water service;
- contributes to customer comfort and premium facility quality;
- can require occasional cleaning/maintenance;
- never creates a per-customer hydration meter.

15.3 Amenity rule

An amenity belongs in the game when it:

- changes customer convenience/quality;
- creates a readable property choice;
- creates manageable operating work;
- helps the facility feel authentic.

It does not belong if it only adds checklist burden.

Water fountains are intentionally light-touch. They make a larger facility feel thoughtful and comfortable but do not create a per-customer thirst simulation.

PART IX - CUSTOMERS

25. Customer Lifecycle and Visible Visit Types

61.1 Customer lifecycle

A customer can move through:

PROSPECT -> LEASED TENANT -> ACCESS VISITS -> BILLING HISTORY -> MOVE-OUT

Possible exception branches:

- service complaint;
- delinquency;
- lien/resolution.

61.2 Prospect evaluation

A prospect evaluates only factors the game actually simulates, such as:

- product availability;
- size;
- climate requirement;
- asking rent;
- market conditions;
- convenience/access if authoritative;
- property quality/security if authoritative.

61.3 Satisfaction

Avoid a giant Sims-style needs panel.

Use a small set of explainable experience dimensions:

- access;
- convenience;
- cleanliness;
- security;
- climate suitability;
- service;
- value.

61.4 Reputation

Reputation may summarize repeated customer experience, but it must remain decomposable.

If reputation is low, the player can inspect why:

- long gate waits;
- dirty hallways;
- repeated elevator delays;
- security weakness;
- poor service.

61.5 Retention / departure

Lease departure pressure can respond to authoritative service/price factors, but no single bad cart trip should instantly evict a tenant.

Use accumulated or periodic evaluation so the player has time to diagnose trends.

61.6 Customer thoughts

Thoughts are sampled explanations of simulation truth, not separate gameplay authority.

Visible visit types

Customer activity should not look like identical cars touching a unit and disappearing.

Useful visit archetypes:

- **Prospect visit:** enters, may visit office/tour, evaluates product, leaves or rents.
- **Move-in:** arrives with more load activity, often uses loading + carts + elevator.
- **Routine access:** short visit to an occupied unit.
- **Large access trip:** more time, greater cart/elevator use.
- **Customer-service visit:** office destination for code/payment/service issue.
- **Move-out:** loading-intensive trip followed by turnover state.
- **Delinquency/resolution interaction:** only when supported by the authoritative lease state.

Visit types change physical behavior, not just animation labels.

26. Customer Conversations

Customer interaction should humanize the simulation without becoming a dialogue RPG.

16.1 Conversation types

Prospect sizing

"I'm storing a two-bedroom apartment. Is a 10x10 enough?"

Access problem

"My gate code isn't working."

Cart request

"Are there any carts available?"

Elevator complaint

"I've been waiting forever for the elevator."

Maintenance complaint

"The light near my unit is out."

Pricing question

"Why did my rate change?"

Move-out

"I'm done with the unit. What do I need to do?"

16.2 Conversation design rule

Every consequential response must map to real game state or policy.

Example - access problem:

- reset/verify code if allowed;
- dispatch Tech if keypad is faulty;
- explain access hours if outside policy;
- deny an invalid request.

Do not create fake dialogue choices whose only result is hidden reputation points.

16.3 Automation at scale

Early game, the player may personally handle some conversations. Later, Clerks and Managers resolve routine requests based on policies, leaving exceptions to the player.

Conversation rule

Conversations are **interfaces into real simulation state**, not a parallel RPG.

A conversation may:

- explain a need;
- expose a problem;
- ask for a real operational decision;
- trigger an authoritative action.

Examples:

- "My gate code isn't working." -> reset access, route to Clerk, or explain a valid account restriction.
- "Do you have any carts?" -> direct the player to the actual cart situation.
- "I need climate control." -> product availability and pricing matter.
- "Why did my rate change?" -> lease/rent policy state matters.

Dialogue should not create fake choices whose outcomes exist only in text.

27. Satisfaction, Reputation, and Retention

Customers are visible agents with a small set of understandable purposes.

17.1 Visit types

- prospect/lease visit;
- move-in;
- routine access;
- move-out;
- office/service visit;
- auction/lien resolution where appropriate.

17.2 Needs

Keep the core needs compact:

- access;
- convenience;
- cleanliness;
- safety/security;
- climate suitability;
- price/value;
- dignity/service.

17.3 Thoughts

Customer thoughts are explanations, not random flavor spam.

Good examples:

- "No carts at Main Loading."
- "That elevator was quick."
- "This hallway is a long walk from loading."
- "Nice covered loading area."
- "Gate line is backing up."

- "The hallway is dark."

Thoughts should point to fixable systems.

Use a small explainable set of experience dimensions:

- access;
- convenience;
- cleanliness;
- security;
- climate suitability;
- service;
- value.

Reputation can summarize those experiences, but the player must be able to inspect the underlying reasons.

PART X - STAFF AND OPERATIONS

28. Owner and Staff Roles

Staff work should not disappear into a progress bar when the work has a physical location.

12.1 Shared task model

Every operational task has:

- task type;
- target;
- urgency/priority;
- eligible roles;
- estimated/authoritative work requirement;
- assigned actor;
- route/access requirement where applicable;
- completion effect.

12.2 Owner

Early game, the Owner can perform basic tasks personally. This teaches what work exists.

12.3 Clerk

Office-facing tasks:

- leasing/service;
- access-code help;
- retail;
- complaints/questions;
- customer-facing administrative work.

12.4 Porter

Property logistics:

- cleaning;
- cart recovery;
- basic make-ready;
- trash/dumpster work;
- simple resets.

12.5 Tech

Equipment maintenance:

- gate/keypad;
- automatic doors;
- lights/cameras;
- elevator;
- HVAC;
- power-related equipment.

12.6 Manager

Later game, the Manager converts individual task clicking into policy-driven operations. The Manager does not make the property irrelevant; they allow the player to focus on higher-level diagnosis, construction and capital decisions.

Core roles

Owner - hands-on early operator.

Clerk - office/customer service, leasing, retail, access help.

Porter - cleaning, cart recovery, make-ready support, property tasks.

Tech - gates, keypads, lighting, cameras, elevators, HVAC, equipment.

Security/Guard - advanced-site security presence where useful.

Manager - later delegation and property-level policy.

Do not add roles merely to create more hiring screens.

29. Hiring, Scheduling, Tasks, and Delegation

Staff Task Lifecycle



Staff task lifecycle: work remains visible on the property.

60.1 Staff record

Each employee has:

- role;
- wage/payroll cost;
- shift/availability;
- current location when actor simulation exists;
- current task;
- assigned area;
- priorities;
- basic work capacity;
- traits only if they create understandable gameplay.

60.2 Hiring

The hiring screen should answer:

- role;
- wage;
- what tasks they can perform;
- work capacity;
- optional simple trait.

Do not require résumé comparison spreadsheets.

60.3 Working hours

Facilities may have office hours and access hours.

Staff shifts can be abstracted into:

- day;
- evening;
- overnight where later needed.

The first facility should avoid shift-scheduling complexity until staffing itself is understood.

60.4 Breaks and needs

Staff needs should remain minimal:

- break;
- restroom;
- manageable walking/workload.

Needs exist to make support rooms and layout matter, not to create a life-sim.

60.5 Task selection

Without direct assignment, staff select eligible work based on:

1. emergency/blocked-service priority;
2. player priority policy;
3. area assignment;
4. route/service cost;
5. age of task.

Exact scoring is implementation detail, but the behavior should be predictable.

60.6 Player control

Early:

- assign this task;
- hire/fire.

Mid:

- role priorities;
- area assignment.

Late:

- Manager policies;
- exception handling.

60.7 Staffing failure

Understaffing appears as:

- waiting customers;
- dirty property;
- unrecovered carts;
- maintenance backlog;
- slow make-ready.

Overstaffing appears primarily as payroll burden and idle staff.

The office as an operational place

The office should have a concrete role:

- leasing/service interactions;
- customer problems;
- retail/locks;
- staff coordination;
- potential actor origin for some staff once true staff pathing exists.

If nobody can provide required office service:

- customers wait;
- online/self-service actions may still function if the property supports them;
- the UI should explain what service is unavailable.

Later automation can reduce office staffing, but it should be an operating strategy.

30. Staff Spatial Logic

The long-term game should model real staff origins and paths when the design proves that added spatial depth is worthwhile.

23.1 Early implementation proxy

A bounded experiment may use service-access distance from existing route authority as a proxy for spatial service burden. It must never be described as literal staff walking if no actor origin exists.

23.2 Target implementation

When full staff pathing exists, an actor has:

- current position;
- allowed movement network;
- assigned role/area;
- current task;
- route to target;
- work duration;
- return/next-task behavior.

23.3 Area assignments

Later properties can use assignments such as:

- Main Building;
- Floors 2-3;
- Exterior Drive-Up;
- Loading/Cart Zone;
- North Wing.

This reduces cross-property walking and creates meaningful staff deployment.

The long-term target is real actor-origin pathing. Any simplified spatial proxy used earlier must remain clearly labeled as a proxy and must not silently become the permanent fiction.

PART XI - CONDITION, MAINTENANCE, AND TURNOVER

31. Maintenance and Equipment Condition

21.1 Wear philosophy

Equipment should usually warn and degrade before severe failure. The player should have time to understand and respond.

21.2 Serviceable objects

Likely service families:

- unit door/lock hardware;
- gate/keypad;
- sliding/automatic doors;
- lights;
- cameras;
- carts;
- elevators;
- HVAC;
- restroom fixtures;
- office/support equipment where meaningful.

21.3 Preventive vs reactive maintenance

Early game: mostly reactive and easy to understand.

Mid game: preventive service becomes available as a policy.

Late game: Manager/Tech staffing can maintain schedules automatically.

21.4 No maintenance chore simulator

The game should generate enough work to make staffing and facility design matter, not so much that every in-game day becomes a list of broken objects.

Maintenance should usually degrade toward a visible warning/problem before catastrophic failure, giving the player time to respond.

32. Cleaning and Property Condition

Cleaning is visible and geographically grounded.

High-traffic areas accumulate more work:

- loading;
- hallways;
- elevator lobbies;
- restrooms;
- office;
- dumpster/cleanout areas.

Owner/Porter work can include:

- sweep/mop;
- trash pickup;
- restroom service;
- spill cleanup;

- cart recovery;
- abandoned-item cleanup.

The player should see dirt/clutter before a dashboard tells them the cleanliness score.

The player should visibly distinguish a clean, well-run facility from a neglected one.

Cleaning sources include:

- high-traffic hallways;
- loading areas;
- restrooms;
- dumpsters/cleanout;
- move-out debris;
- occasional spills/obstructions.

33. Turnover and Make-Ready

MOVE-OUT -> UNIT UNREADY -> INSPECT -> CLEAN/REPAIR -> RENT-READY -> RE-LEASE

Turnover is where the physical property, labor and economics meet most clearly.

A distant or awkward unit can require more service effort once spatial work is authoritative. A broken roll-up door can extend the service chain. A backlog leaves capacity physically unavailable and economically idle.

The canonical loop is:

MOVE-OUT -> NOT RENT-READY -> INSPECT -> CLEAN/REPAIR -> RENT-READY -> RE-LEASE

This is one of the best places for layout, staff capacity, and service distance to become economic consequence.

PART XII - MARKET, LEASING, AND MONEY

34. Market, Demand, Pricing, and Leasing

Construction is not enough; the property must still be a business.

24.1 Demand

Demand is by meaningful product class: size, environment and later property/market distinctions. Demand is not a literal queue of guaranteed tenants.

24.2 Asking rent

Player sets asking rent by product group. Rent influences move-in response but does not create a single "correct" number.

24.3 Lost prospect reasons

When a prospect does not rent, the game should explain plausible authoritative reasons where available:

- price too high relative to market;
- no suitable size;
- no climate availability;
- access/convenience friction if that coupling exists;
- no rent-ready inventory.

24.4 Billing truth

Monthly billing and daily operating costs must be presented on their real time scales. A seven-day window with no billing event can be negative while the business remains healthy.

Demand should be a market signal, not a literal queue of guaranteed customers.

Different markets should create different viable strategies rather than one universal optimal unit mix.

35. Billing and Financial Presentation

27.1 Core financial questions

The player should always be able to answer:

- How much cash do I have?
- What rent is contracted monthly?
- What rent was actually collected recently?
- What does the property cost to operate?
- What did I spend on construction?
- Is this facility producing enough cash to support the next improvement?

27.2 Cost families

- base property operating cost;
- payroll;
- electricity;
- HVAC;
- elevator operation/maintenance;
- water/amenities;
- cleaning/maintenance;

- security equipment;
- construction/capital spending.

27.3 Capital tradeoffs

A premium interior building may require more capex but produce greater density and product quality. A drive-up strip may be cheaper/simple but consume more land. The game should create these kinds of understandable choices.

The game must distinguish:

- contract/monthly rent roll;
- actual collected rent;
- daily/period operating cost;
- payroll;
- capital construction;
- current cash.

A short negative 7-day window is not automatically a failing property if billing days fall outside that window.

36. Delinquency, Liens, and Auctions

Keep the system self-storage-specific but approachable.

Stages:

- Current;
- Past Due;
- Delinquent;
- Lien/Resolution Eligible;
- Resolution in Progress;
- Auction/Other Resolution.

The player should manage policy and exceptions, not perform legal paperwork line by line.

At larger properties, auction day can become a visible set piece using an auction/service court.

This system should feel specific to self-storage while remaining readable and operational rather than legal-paperwork-heavy.

37. Capital, Financing, and Investment

The game should be a business simulation, but financing should not drown the property game.

63.1 Core operating statement

For each property, the player can understand:

$$\text{COLLECTED RENT} + \text{ANCILLARY REVENUE} - \text{OPERATING COST} - \text{PAYROLL} - \text{SERVICE COST} = \text{OPERATING CASH CONTRIBUTION}$$

Capital construction is shown separately so a profitable property does not look "unprofitable" merely because the owner just built a new wing.

63.2 Capex vs operating expense

Player-facing financial views distinguish:

- construction/capital spending;
- recurring operating expense;
- collected revenue;
- contractual rent roll.

63.3 Financing progression

Early game:

- mostly cash-funded improvements to teach fundamentals.

Mid/late game:

- optional debt/financing can support larger acquisitions or construction.

A simple model is preferred:

- principal;
- payment;
- interest rate;
- term or simplified monthly obligation;
- borrowing limit/approval.

Do not turn the game into commercial mortgage underwriting.

63.4 Capital allocation

At company scale, the player chooses among:

- expansion at Property A;
- elevator/HVAC improvement at Property B;
- acquisition/development of Property C;
- reserve cash.

This is where "tycoon" becomes real.

63.5 No hidden solvency trap

Before a major build, show:

- current cash;
- project cost;
- expected recurring operating cost;
- resulting cash reserve.

Do not promise future revenue.

Capital decisions should eventually include:

- add units;
- improve loading;
- add elevator capacity;
- modernize security;
- add HVAC;
- buy another parcel/property;
- keep reserve cash.

The game should show the immediate financial consequence before a major investment without pretending to forecast guaranteed future revenue.

PART XIII - OPERATING THE FACILITY

38. Operating Hours, Access Hours, Day/Night, and Weather

These systems add authenticity only when kept readable.

67.1 Office hours

The office can have service hours. Outside office hours:

- Clerk service is unavailable unless staffed;
- tenant gate access may continue according to access policy.

67.2 Tenant access hours

Policies can progress from:

- standard access window;
- extended access;
- 24-hour access where property/market supports it.

Access hours can affect convenience and security workload.

67.3 Day/night

Lighting and security are most visually meaningful with day/night presentation.

Time of day should affect:

- light visibility;
- staff shift presence later;
- office availability;
- customer arrival distribution.

It should not create twitch pressure.

67.4 Weather

Weather can justify covered loading and indoor comfort.

A restrained model can include:

- fair;
- rain;
- heat/cold condition where climate relevance is useful.

Weather should modify convenience, not repeatedly punish the player with random disasters.

Covered loading becomes visibly valuable during rain without becoming mandatory in every climate.

Weather should be restrained. Its main value is making design choices such as covered loading and indoor climate visibly useful, not generating random punishment.

39. Failure, Recovery, Win, and Loss

Every common problem should have a readable cause and at least one practical recovery path.

Problem	What player sees	Likely causes	Responses
No carts at loading	empty corral, customers searching/waiting	too few carts, poor return pattern, bad corral placement	buy carts, add/move corral, recovery priority
Elevator queue	people/carts waiting	too much upper-floor demand, slow/poor-condition elevator	service, upgrade, add elevator, redesign future capacity
Gate backup	vehicle queue	slow keypad/gate, one access point, traffic concentration	repair/upgrade, add capacity if supported, change layout

Problem	What player sees	Likely causes	Responses
Dark hallway	visible darkness	missing/broken lighting	add/repair lights
Climate zone overloaded	warning at plant/zone	too much climate area for capacity	add HVAC, reduce/stage zone, repair equipment
Service backlog	pending icons/tasks	too few staff, long routes, too much wear	hire, reprioritize, redesign, preventive maintenance
Cash crunch after expansion	low cash, construction spend visible	overbuilding, weak leasing, high operating burden	pause expansion, adjust price, operate to recover

No hidden punishment should be required for these to matter.

The player must be able to make a bad facility and learn from it.

The game should usually warn before a decision becomes catastrophic:

- route is long;
- elevator demand is high;
- power capacity is near limit;
- HVAC zone is overloaded;
- cart stock is frequently low;
- staff backlog is rising;
- construction will leave little cash.

Warnings explain facts, not recommendations. The player decides.

Can the player lose?

Normal sandbox and campaign play should favor **recoverable financial distress** over sudden hidden game-over.

Possible severe failure path:

1. cash reserve falls;
2. player receives clear financial warning;
3. expansion becomes difficult;
4. the player can cut costs, change rents, sell/close capacity where supported, obtain financing later, or restructure;
5. only sustained insolvency with no valid recovery can trigger property/company failure if the selected mode permits it.

Scenario Mode may define explicit fail conditions, but they must be visible from the start.

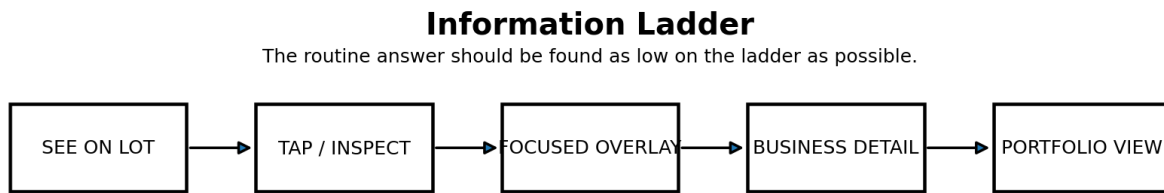
What counts as winning?

There is no single universal end screen.

- **Tutorial:** graduate the property.
- **Scenario:** satisfy the authored completion conditions.
- **Sandbox:** no mandatory ending.
- **Company/Campaign:** reach company milestones while continuing play if desired.

The player's long-term reward is mastery and scale, not a countdown victory.

40. Diagnostics and Information Ladder



Information ladder: lot first, detail only when needed.

29.1 Level 1 - physical tell

See:

- queue;
- empty cart corral;
- dark hallway;
- maintenance worker;
- broken door;
- full loading bays.

29.2 Level 2 - inspection

Tap the object. Answer:

1. What is this?
2. Is it working?
3. Why?
4. What can I do?

29.3 Level 3 - focused overlay

Examples:

- access/traffic;
- cart distribution;
- elevator/loading congestion;
- security/lighting;
- HVAC coverage;
- staff workload;
- occupancy/demand.

29.4 Level 4 - business detail

Use when comparing trends, costs, pricing and facility performance.

29.5 Level 5 - portfolio

Only after multiple properties exist.

The information ladder is:

1. SEE IT ON THE LOT

2. TAP / INSPECT IT

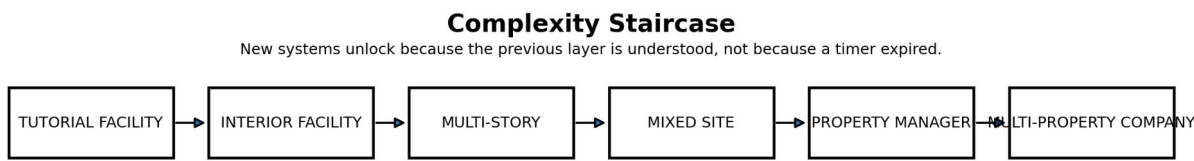
3. OPEN A FOCUSED OVERLAY

4. OPEN DETAILED BUSINESS / OPERATIONS DATA

The game should not force Level 4 for ordinary diagnosis.

PART XIV - PROGRESSION

41. Complexity Staircase



Complexity staircase from beginner property to company management.

New mechanics appear because the player encounters a problem the mechanic solves.

Stage	New problem	Mechanic introduced	Mastery signal
Tutorial	vacancy needs work	make-ready/Owner work	player restores unit to rent-ready
Tutorial	site reaches capacity	construction	player adds/commissions capacity
Tutorial	interior move-in needs equipment	carts/loading	player understands cart availability
Tutorial	Owner becomes bottleneck	employee	player delegates successfully
Early	larger indoor wing needs control	cameras/keypads/lights	player fixes coverage/access issue
Early-mid	market supports premium environment	HVAC/climate	player commissions viable climate capacity
Mid	land constraint	second floor/elevator	player designs functional vertical flow
Mid	bigger site creates work geography	staff areas/priorities	player reduces backlog through management
Late	routine work distracts from strategy	Manager	property runs under policies
Late	one property is stable	second property	player allocates capital across sites

42. Unlocking, Research, and Mastery

Unlocking should feel like learning the storage business rather than filling an XP bar.

64.1 Three unlock sources

Problem-earned

A new mechanic becomes available after the player encounters the problem it solves.

Property-earned

Certain facility types introduce systems appropriate to that property.

Company-earned

Scale unlocks delegation, capital tools, and advanced upgrades.

64.2 Example progression

Maple Street:

- drive-up;
- basic indoor;

- carts;
- basic keypad/security;
- Owner/first employee.

Urban property:

- covered dock;
- automatic wide doors;
- multiple corrals;
- elevators;
- climate;
- stronger access/security;
- bathroom/water amenities;
- Tech specialization.

Large mixed property:

- multiple traffic zones;
- vehicle/RV;
- multiple staff areas;
- auction/service space;
- advanced security;
- larger capital decisions.

Company:

- Managers;
- acquisitions;
- standardized upgrade programs;
- cross-property capital allocation.

64.3 Mastery gates

Do not require subjective "play well enough" scoring for tutorial continuation.

Use deterministic state milestones such as:

- first unit restored to rent-ready;
- first lease after turnover;
- first expansion commissioned;
- first successful cart-supported interior trip;
- first delegated task completed;
- first diagnosed/repaired service fault;
- first viable climate area commissioned;
- first functional upper floor opened.

Progression philosophy

The game should avoid a generic "research points unlock elevator" tree unless a true research system later proves useful.

Preferred unlock logic:

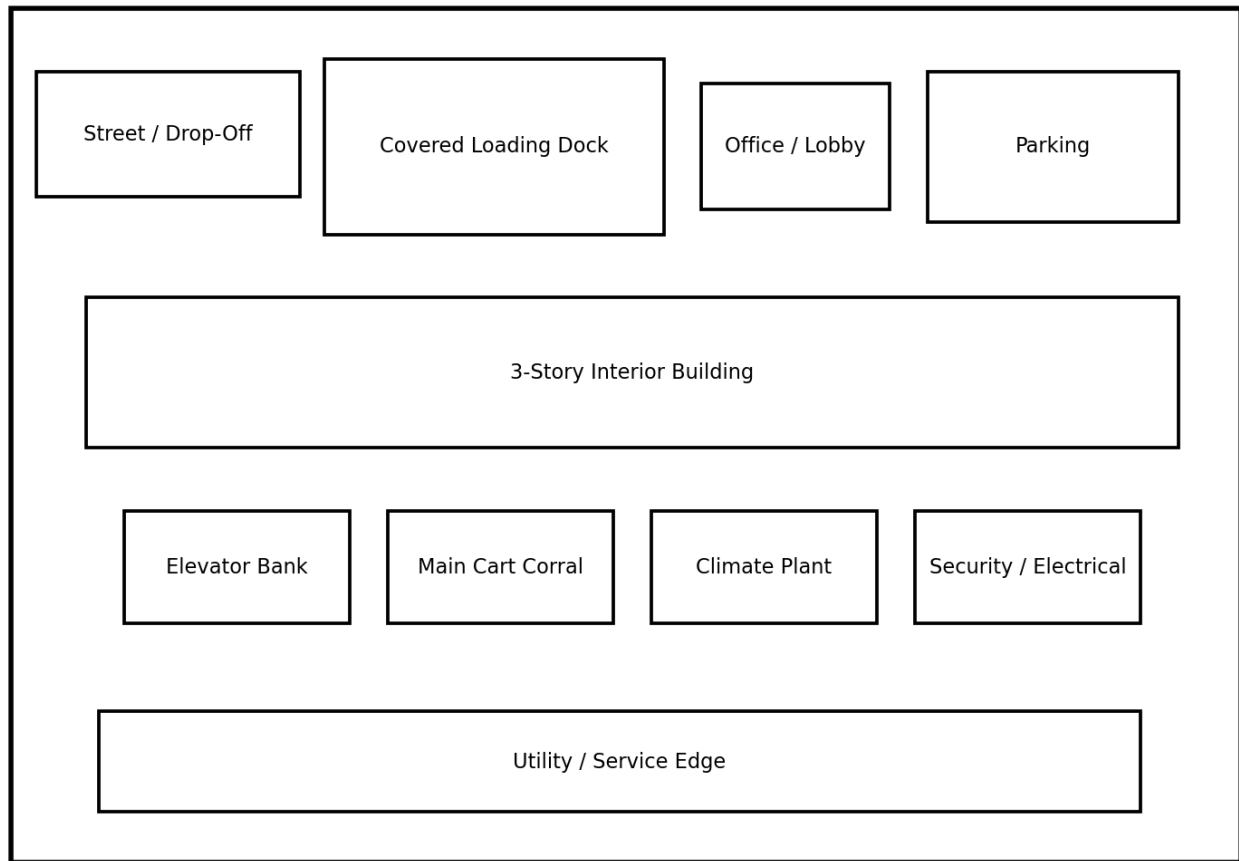
- **problem-earned:** player encounters the need;
- **property-earned:** a facility archetype introduces the system;
- **company-earned:** scale unlocks management/capital tools;
- **vendor/market availability:** advanced equipment becomes purchasable through company progression.

An elevator should feel like a capability the company becomes ready to deploy, not a magical technology the player invents.

PART XV - PROPERTIES AND SCENARIOS

43. Canonical Property Archetypes

Property Archetype 2: Multi-Story Urban Storage



Expensive land forces vertical density; logistics, climate, access and elevator capacity become the design game.

Urban multi-story property archetype.

43.1 Maple Street - Beginner Owner-Operator

See Section 5. Purpose: teach the game on a small, understandable property with drive-up and a modest interior building.

Player identity: **owner-operator**.

Main lessons:

- basic access;
- make-ready;
- rent;
- expansion;
- carts;
- basic maintenance;
- first delegation.

43.2 Multi-Story Urban Storage

A small/expensive parcel makes vertical density the obvious opportunity.

32.1 Defining systems

- covered loading dock;
- wide automatic/sliding doors;
- dense interior units;
- elevators and stairs;
- multiple cart corrals;
- climate control;
- camera/light coverage;
- building keypads;
- bathrooms and water fountain;
- specialized Porter/Tech work.

32.2 Player questions

- How much upper-floor capacity can one elevator support?
- Where should the cart corral be relative to loading and elevators?
- How wide should the main hallway be?
- Is another elevator worth losing rentable floor area?
- Which floors/wings should be climate-controlled?
- Can one loading dock support the building at peak activity?

This property should prove that indoor/multi-story design is fun enough to carry the game's middle.

43.3 Large Mixed Facility

This facility combines drive-up, interior, vehicle storage and more complex operations.

33.1 Defining systems

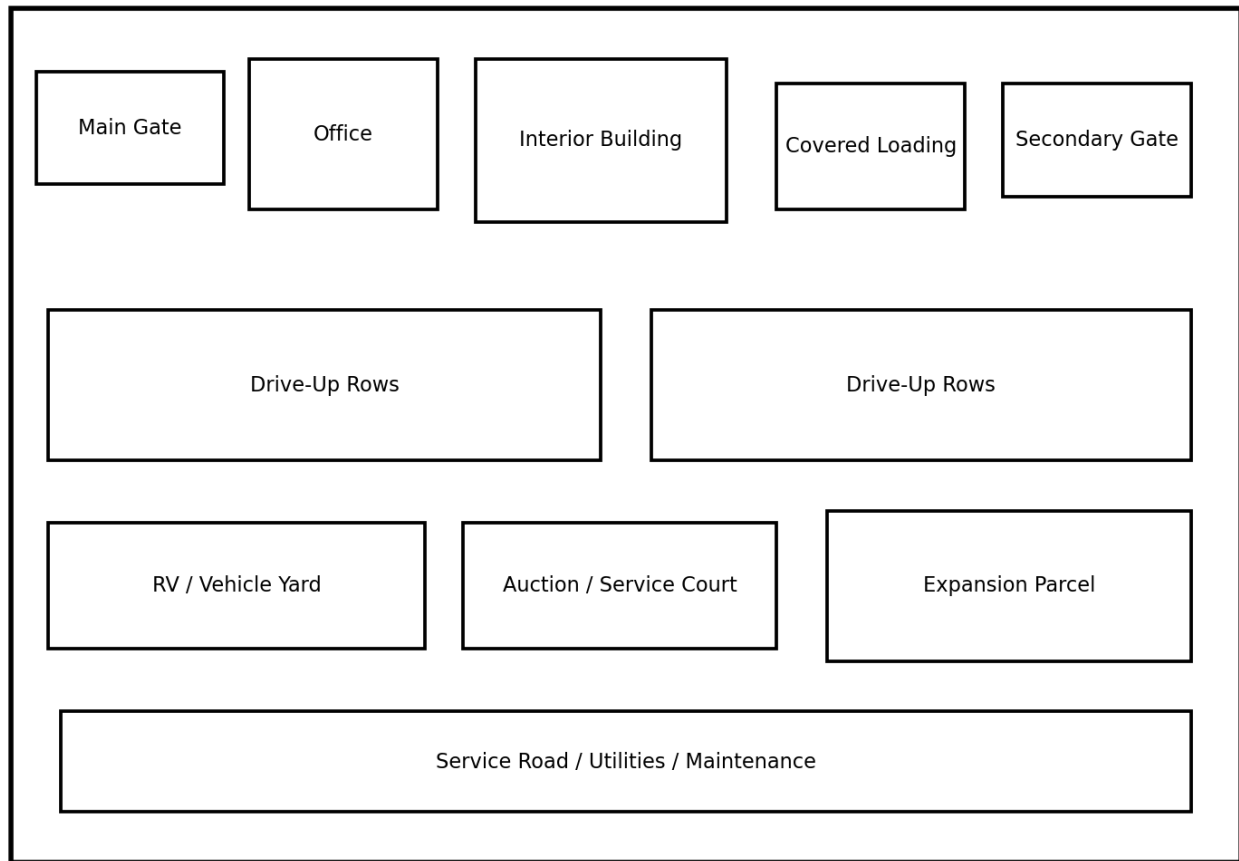
- multiple gates/traffic flows;
- drive-up rows;
- interior building;
- covered loading;
- vehicle/RV area;
- auction/service court;
- larger staff;
- multiple maintenance/security zones;
- expansion parcel.

33.2 Player questions

- Which customer types deserve the best frontage?
- Should a new building use the existing gate or get its own entrance?
- Where should staff/service facilities sit to minimize cross-site work?
- Which product type should receive the next capital investment?
- When is it better to hire than to redesign?

44. Scenario Design Contract

Property Archetype 3: Large Mixed Facility



The player now integrates multiple product types, traffic patterns, staff zones and capital choices.

Large mixed-property archetype used for scenario variety.

Every scenario must define:

- **Fantasy:** what kind of storage problem is this?
- **Starting property:** what physically exists?
- **Market:** what demand/price context exists?
- **Starting finances:** what resources and liabilities exist?
- **Known problems:** what the player can see at start.
- **Hidden-but-discoverable problems:** only when fair and diagnosable.
- **Available systems:** what is unlocked.
- **Completion conditions:** state-based, visible, non-arbitrary.
- **Failure conditions:** if any, visible before play.
- **Optional mastery goals:** extra challenge, never required for basic completion.

Scenario goals should support multiple valid solutions whenever possible.

Example: "Reduce average loading/elevator delay below X and sustain positive operating contribution for Y days" is better than "Build exactly two elevators."

PART XVI - SANDBOX

45. Sandbox Configuration

45.1 Standard Sandbox

Player chooses:

- empty lot or existing facility;
- market profile;
- parcel;
- starting capital/difficulty;
- tutorial hints on/off;
- incident frequency;
- optional advanced systems;
- company progression on/off where supported.

The default should be curated enough that the player can start quickly.

45.2 Creative Sandbox

Optional toggles:

- unlimited money;
- full catalog unlocked;
- instant construction;
- maintenance off;
- incidents off;
- demand normal or relaxed;
- full customer/staff/logistics simulation retained.

45.3 Generated existing-facility starts

A generated property should never be random nonsense.

Generation rules should produce:

- a valid functioning facility;
- recognizable building/road hierarchy;
- deliberate strengths;
- a small number of plausible weaknesses;
- a coherent unit mix for its market.

Generated facilities can become excellent long-term replayability content once the authored archetypes prove the rules.

PART XVII - COMPANY / TYCOON LAYER

46. Multi-Property Company Architecture

46.1 Property ownership

The company can eventually:

- develop raw land;
- acquire existing facilities;
- sell properties if supported;
- renovate/reposition acquired properties.

46.2 Money

Each property has its own operating statement.

Company-level cash/capital may be pooled or transferable, but the UI must preserve property-level truth so a weak property cannot hide permanently behind a successful one.

46.3 Managers

A Manager can automate:

- routine task assignment;
- staffing priorities;
- ordinary maintenance scheduling;
- approved rent-policy bands where later supported;
- notifications/escalation.

The Manager should not erase the property. The player can always visit and inspect what is happening.

46.4 Time across properties

Target architecture:

- all owned properties advance under the same company simulation clock;
- the player may pause the entire company;
- properties continue routine operations when not currently viewed;
- unresolved exceptions surface through a company-level notification/attention system.

The player should not need to individually "run" each property in parallel to prevent decay.

46.5 Company-level decisions

At scale, the player chooses:

- where to invest;
- which property to expand;
- which property to modernize;
- where to hire/replace management;
- whether to acquire/develop;
- how much cash to reserve.

This is where the title earns **Tycoon**.

PART XVIII - INTERFACE AND PRESENTATION

47. Primary Modes

Build

34.1 Entering Build

Time does not need to stop automatically, but the player can pause instantly. The default tutorial behavior should strongly encourage pausing during construction without requiring it.

34.2 Build palette

Recommended categories:

- Site;
- Roads & Loading;
- Buildings;
- Units;
- Interior;
- Access & Security;
- Utilities;
- Customer Amenities;
- Staff & Service;
- Scenery/Wayfinding.

34.3 Live preview

Preview shows:

- valid/invalid footprint;
- route/access requirement;
- doors/frontage;
- coverage/dependency warnings;
- cost;
- construction time/phase where authoritative;
- ongoing operating cost where useful.

34.4 Confirmation

The player confirms once. The simulation commits atomically: money and construction state cannot disagree.

Operate

Operate is for the property right now.

Primary uses:

- inspect customers/staff/tasks;
- manage Owner/staff assignments;
- start maintenance/make-ready;
- inspect carts/elevators/loading;
- respond to access/customer issues;
- use operations-health overlays.

The lot remains visible behind all controls.

Business

Business answers:

- What am I charging?
- What is occupied?
- What is collecting?
- What are my costs?
- What is hurting leasing/retention?

It should not become a permanent spreadsheet screen. Tapping a business problem should return the player to the relevant physical place when possible.

Growth

Growth answers:

- Where is capacity constrained?
- What products have market opportunity?
- What can I afford?
- What does the next expansion cost to build and operate?
- What physical space is available?

Growth should point back toward the property rather than provide a magic "expand" button.

These are lenses on the same facility, not four separate games.

48. iPhone Interaction Contract

66.1 Property-first screen budget

The lot should occupy the clear majority of the screen during ordinary play.

Persistent chrome should be restricted to:

- money/key status;
- mode;
- time controls;
- selected-object/action affordance.

66.2 Bottom sheets

Detailed inspection uses a bottom sheet or compact panel that can:

- peek;
- half-expand;
- full-expand.

The player can dismiss it without losing property context.

66.3 Build palette

The palette should support:

- large touch targets;
- category tabs;
- recently used/favorites later;
- long-press detail;
- no desktop hover dependency.

66.4 Precision placement

Provide:

- grid snapping;
- alignment guides;
- orientation ghost;
- optional nudge arrows for fine placement;
- visible door/frontage direction.

66.5 Error messages

Never use only color.

Bad:

- red tile.

Good:

- red footprint + "Blocked: overlaps hallway."
- red door marker + "Unit door needs hallway frontage."

66.6 Floor readability

The active floor must be obvious at all times. Hidden floors must not intercept taps.

66.7 Selection priority

When objects overlap visually, the game should prefer the most likely interactive layer and allow cycling/context selection when ambiguous.

66.8 One-handed reality

Frequent actions should sit within reachable screen regions where practical. Critical time controls must remain available during menus and inspection.

The property must remain the dominant visual object on the screen.

49. Art Direction Bible

49.1 Canonical visual north star - LOCKED

The following wording is the canonical visual target and should be preserved verbatim in future art briefs:

Warm, readable, stylized 3D realism with miniature-diorama clarity.

Real self-storage architecture, equipment, vehicles, people, and operations simplified into strong readable forms, attractive materials, slightly exaggerated proportions, and expressive animation.

Planet Coaster charm + RollerCoaster Tycoon readability + Prison Architect systemic clarity - grounded specifically in contemporary American self-storage.

This is the standard against which environments, characters, vehicles, props, UI, animation, VFX, lighting, and concept art are judged.

The game must not drift into photorealism, flat sprite presentation, generic mobile-game cartoon art, childish toy styling, sci-fi/futuristic design, sterile architectural visualization, or noisy theme-park spectacle.

49.2 Visual identity in one sentence

Self Storage Tycoon should look like a beautiful, tactile, living miniature model of a believable American self-storage property whose operations remain instantly readable on an iPhone.

49.3 Approved primary visual-reference set



Approved Visual Reference A - modern suburban mixed facility. Visual direction only; layout and branding are not canon.



Approved Visual Reference B - dense urban multi-story facility. Visual direction only; layout and branding are not canon.

Two concept images are approved as primary visual-direction references:

Reference A - Modern suburban mixed facility

Self Storage Tycoon - Master GDD v1.1 - Art + Audio Complete Target Specification

- establishes the broad property camera;
- office near entrance;
- gate/keypad/fence;
- exterior drive-up units;
- interior-storage building;
- covered loading;
- carts;
- seamless exterior/interior cutaway;
- vehicles, customers, staff, HVAC and security equipment;
- restrained materials and warm diorama presentation.

Reference B - Dense urban multi-story facility

- proves the same style can support vertical storage;
- multiple floors in readable cutaway;
- covered loading bays;
- carts and active customers;
- aligned elevators/stairs;
- upper-floor storage corridors;
- staff and vehicle activity;
- rooftop HVAC and modern urban storage architecture.

These images are **visual-direction evidence**, not literal layout, branding, geometry, balance, or implementation authority.

The exact architecture, colors, names, unit counts, floor plans, or branding in a concept image do not become canon merely because the image is approved.

49.4 Camera and projection

The primary gameplay camera is a **controlled elevated three-quarter / near-isometric 3D view**.

Goals:

- property layout reads quickly;
- building fronts and roofs are understandable;
- roads and movement paths remain visible;
- interiors can transition naturally into cutaway;
- camera does not require free-flight 3D skill.

Target interaction:

- one-finger pan;
- pinch zoom;
- limited rotation only if it improves comprehension;
- focus selected object;
- floor selection for multi-story properties.

The camera should feel closer to operating a premium architectural miniature than controlling a first-person camera.

49.5 Scale and proportion rules

Physical dimensions should remain believable, but gameplay-critical objects may be modestly exaggerated for readability.

Allowed exaggeration:

- roll-up doors slightly clearer/larger;
- carts chunkier and easier to recognize;
- keypad pedestals more visible;

- security cameras slightly more pronounced;
- elevator doors and lobbies more distinct;
- staff tools more readable;
- unit-number signage larger than strict real-world scale;
- door-state and maintenance-state animation slightly amplified.

Do not distort:

- building massing;
- vehicle-to-building scale;
- hallway logic;
- elevator placement;
- loading-bay scale;
- human height to the point that the facility becomes toy-like.

49.6 Miniature-diorama treatment

The world should feel tactile and hand-composed:

- clean silhouettes;
- controlled material complexity;
- readable edge definition;
- restrained surface wear;
- soft but clear lighting;
- strong object grouping;
- subtle ambient detail.

The goal is **miniature clarity**, not literal plastic-toy rendering.

At wide zoom, the whole property should read as a board.

At medium zoom, operations should become legible.

At close zoom, individual staff/customer actions and equipment should become charming.

49.7 Contemporary American self-storage architecture

The visual library should be grounded in recognizable real-world storage architecture.

Common exterior vocabulary:

- corrugated or architectural metal panel;
- painted concrete;
- stucco office frontage;
- asphalt;
- roll-up doors;
- chain-link or ornamental security fencing;
- yellow bollards/curbs;
- commercial glass doors;
- metal canopies;
- keypad pedestals;
- monument signs;
- security cameras;
- rooftop HVAC;
- loading aprons;
- box-truck-compatible clearances.

Common interior vocabulary:

- painted/sealed concrete floors;
- bright corridor lighting;

- roll-up interior unit doors;
- unit-number plaques;
- controlled-access doors;
- elevator lobbies;
- wide loading entrances;
- cameras;
- wayfinding;
- carts/corrals;
- HVAC grilles/equipment cues;
- restrained safety markings.

A screenshot without the logo should still unmistakably read as self-storage.

49.8 Property age and archetype variation

The same visual language must support different property personalities.

Older suburban

- faded but readable metal;
- patched asphalt;
- simple office;
- chain-link perimeter;
- older fixtures;
- visible deferred maintenance;
- less polished signage.

Modern suburban

- clean façade;
- coordinated trim;
- landscaped entrance;
- improved lighting;
- decorative fencing;
- covered loading;
- stronger wayfinding.

Dense urban multi-story

- taller façade;
- compact loading apron;
- large automatic doors;
- elevator-forward circulation;
- climate-controlled interior;
- strong cutaway readability.

Large mixed facility

- drive-up rows;
- interior building;
- vehicle/RV area;
- multiple circulation zones;
- more service/staff infrastructure.

Premium facility

- refined canopy;
- cleaner architectural detailing;
- stronger landscaping;
- polished lobby/loading;

- excellent wayfinding;
- integrated security and climate presentation.

The game should communicate improvement without turning every upgraded property into luxury architecture.

49.9 Exterior storage-unit language

Drive-up units should communicate:

- size;
- door orientation;
- occupied/vacant/service state where appropriate;
- row continuity;
- vehicle approach;
- unit number.

Use:

- strong roll-up-door rhythm;
- readable unit numbers;
- simple color families;
- clean wall/roof forms;
- recognizable gutters/bollards only where they add realism.

Avoid excessive tiny detail that destroys row readability.

49.10 Interior storage-unit and hallway language

Interior corridors are one of the game's signature visual spaces.

They should feature:

- repeated but not monotonous unit doors;
- clear unit numbers;
- readable main-vs-secondary corridor hierarchy;
- bright practical lighting;
- clean floor materials;
- visible cameras/keypads;
- elevator/stair landmarks;
- wayfinding signs;
- cart-compatible geometry.

Main halls and elevator/loading lobbies should visually read as higher-capacity spaces.

49.11 Cutaway and floor presentation

Interior cutaway must feel like opening a dollhouse, not switching to a blueprint.

Rules:

- selected roof lifts/fades cleanly;
- floors above hide or ghost;
- camera-facing walls may cut away;
- selected floor remains fully rendered;
- elevator/stair shafts remain vertically legible;
- hidden floors never intercept touches;
- exterior context remains visible enough to preserve orientation.

For multi-floor buildings, vertical alignment should be visually obvious.

49.12 Loading infrastructure

Loading must be visually prominent because it is a major gameplay bridge.

Assets:

- uncovered loading area;
- covered loading canopy;
- full covered loading dock;
- numbered bays;
- automatic/wide sliding doors;
- bollards;
- cart corrals;
- loading signs;
- truck-safe apron markings.

Loading areas should visually communicate whether they are basic, crowded, efficient, premium, or under construction.

49.13 Carts and cart corrals

Carts are a signature Self Storage Tycoon object.

The standard cart silhouette should be recognizable at gameplay zoom:

- low platform;
- upright handle;
- four visible caster/wheel cues;
- robust metal/rubber construction;
- slightly exaggerated thickness and handle height.

Cart corrals should make inventory obvious.

The player should be able to visually notice:

- empty corral;
- healthy stock;
- cart in use;
- stranded cart;
- damaged/out-of-service cart;
- Porter recovering carts.

Do not use a generic airport-luggage-cart silhouette if it weakens self-storage identity.

49.14 Elevators and stairs

Elevator doors must remain obvious in dense cutaways.

Visual requirements:

- distinctive door frame;
- floor indicator;
- queue area;
- enough lobby space to read carts and waiting customers;
- service/maintenance state;
- aligned vertical shaft.

Stairs should be visually readable as human-access redundancy without competing with elevators as the primary moving-goods system.

49.15 Office and customer-service spaces

The office should feel welcoming but operational:

- clear public entrance;
- service counter;
- lock/box retail cues;

- visible staff;
- glass/door frontage where appropriate;
- branded but restrained signage;
- customer parking relationship.

Avoid turning the office into a generic retail store or luxury hotel lobby.

49.16 Security, access, lighting, and utilities

Keypads

- readable pedestal/wall silhouette;
- clear interaction height;
- restrained status light;
- visible enough for customer animation.

Cameras

- recognizable fixed/dome/PTZ families if needed;
- slightly exaggerated silhouette;
- placed where coverage makes sense.

Lighting

- practical commercial fixtures;
- visible pools of light at night;
- no neon theme-park glow.

HVAC

- believable rooftop or service equipment;
- simplified forms;
- fan movement/state where useful;
- enough visual identity to support climate-control gameplay.

Power/service equipment

- readable large equipment only;
- no wire-level visual clutter.

49.17 Bathrooms, water fountains, and amenities

Amenities should read as believable commercial-facility components.

Water fountains:

- simple wall or bottle-fill/fountain form;
- readable near lobby/restroom zones;
- never oversized into a gameplay centerpiece.

Bathrooms:

- clear doorway/signage;
- no unnecessary interior detail unless selected/inspected.

Benches, waiting areas, and small convenience features should remain secondary to storage operations.

49.18 Customers

Customers are slightly stylized adults with believable clothing and body variety.

At gameplay distance, communicate through:

- silhouette;
- carried objects;
- cart use;

- vehicle;
- pose;
- route;
- visit type.

Useful visible customer archetypes:

- prospective renter;
- person carrying boxes;
- couple moving furniture;
- small-business customer;
- family mover;
- routine-access tenant;
- customer with cart;
- customer waiting for service.

Avoid:

- giant heads;
- chibi proportions;
- excessive facial comedy;
- character art that visually overpowers the property.

49.19 Staff-role silhouettes

Roles should be readable without permanent floating labels.

Owner

- neutral practical work appearance.

Clerk

- office/service attire;
- clipboard/tablet/counter context.

Porter

- work shirt/gloves;
- cleaning/cart-recovery equipment.

Tech

- tool bag;
- technical workwear;
- visible equipment interaction.

Security

- distinct but non-militarized.

Manager

- professional-casual;
- less task-equipment-driven.

The strongest role cue should be **what the person is physically doing**.

49.20 Vehicles

Core vehicle families:

- sedan;
- SUV;
- pickup;
- cargo van;

- small box truck;
- large moving truck;
- RV/trailer later.

Vehicle forms should be simplified for readability but retain believable turning footprints and scale.

Visit type should alter behavior:

- routine tenant visit;
- office visit;
- drive-up access;
- loading-bay move-in;
- move-out;
- service/vendor visit.

49.21 Construction stages

Large construction should visually progress.

Target stages:

1. marked footprint / site prep;
2. slab/foundation;
3. structure/shell;
4. enclosure/exterior skin;
5. interior fit-out;
6. doors/equipment;
7. complete but not commissioned;
8. operating.

Small installs can use abbreviated stages.

Construction should be satisfying to watch but must not create false gameplay state.

49.22 Maintenance, cleaning, and operational imperfection

The world should not look like a permanent real-estate brochure.

Visible truthful imperfections can include:

- cart left upstairs;
- dirty/loading-area patch;
- light out;
- keypad being serviced;
- HVAC maintenance;
- unit make-ready;
- blocked area;
- active construction;
- customer queue;
- worn asphalt;
- older paint;
- temporary service sign.

These provide gameplay stories without making the facility grim or filthy.

49.23 Animation philosophy

Animation communicates simulation.

High-value animations:

- gate opens/closes;
- keypad entry;
- car queues and parks;
- roll-up door opens;
- sliding door opens;
- customer takes/returns cart;
- cart rolls;
- elevator arrives/doors open;
- customer enters/exits elevator;
- Porter recovers cart;
- staff sweeps/cleans;
- Tech repairs equipment;
- Owner performs make-ready;
- construction activity;
- light activates;
- HVAC fan runs.

Animation can be expressive but must remain physically believable.

Do not play an operational animation unless the corresponding authoritative activity is actually happening.

49.24 VFX philosophy

VFX are restrained and informational.

Good:

- subtle dust during construction;
- small sparks during repair;
- rain splash;
- light steam/airflow hint where appropriate;
- selection outline;
- route ghost;
- subtle completion highlight.

Avoid:

- coin explosions;
- confetti for ordinary leases;
- glowing buildings;
- giant stars;
- constant particles;
- neon upgrade arrows;
- effects that obscure the property.

49.25 Color hierarchy

World color:

- believable asphalt;
- concrete;
- galvanized/painted metal;
- restrained branded accents;
- warm office materials;
- clean interior whites/grays with controlled color.

Gameplay state:

- green: working/available/connected;
- amber: degraded/attention;

- red: blocked/failure;
- blue/cyan: climate/utility/service coverage;
- dedicated security/access color only if needed.

Do not paint the normal world like a debug heat map.

49.26 Lighting, day/night, and weather art

Day

- warm/neutral daylight;
- readable shadows;
- clear materials.

Evening

- office glow;
- practical site lighting;
- more visible security fixtures.

Night

- readable rather than cinematic darkness;
- lighting coverage matters;
- no loss of gameplay readability.

Rain

- wet asphalt;
- canopy usefulness;
- subtle reflections;
- readable customer/loading behavior.

Weather should enrich the miniature world without overwhelming it.

49.27 UI and iconography

UI should feel like a modern self-storage operating company.

Target language:

- clean industrial;
- off-white/light panels;
- charcoal/navy structure;
- restrained safety-yellow or brand accents;
- simple line/solid icons;
- strong hierarchy;
- minimal decorative chrome.

Avoid:

- giant generic mobile-game buttons;
- treasure chests;
- currency particle art;
- sci-fi panels;
- faux wood;
- gratuitous gradients;
- red-badge overload.

The property should remain the hero.

49.28 Typography and signage

World signage:

- legible commercial sans-serif;
- unit numbers readable at gameplay distance;
- consistent wayfinding hierarchy;
- believable American storage branding.

UI typography:

- high readability on iPhone;
- clear numerical hierarchy;
- no decorative type for operational data.

49.29 Overlays and diagnostics

Overlays should reveal one system at a time.

They may use:

- transparent coverage;
- route lines;
- queue indicators;
- soft floor tint;
- icons;
- object outlines.

They must not permanently recolor the world.

The player should be able to return immediately to the attractive normal property view.

49.30 Asset modularity

Art production should favor reusable modular kits.

Examples:

- roll-up-door modules;
- unit wall segments;
- hallway corners/intersections;
- office façade pieces;
- canopy segments;
- fence/gate modules;
- parking markings;
- loading equipment;
- elevator/stair modules;
- camera/light/keypad families;
- HVAC modules;
- cart families;
- vehicle families.

Modularity should create variety through combination, material, age, signage, and layout - not endless bespoke meshes.

49.31 Performance-oriented art rules

The visual target must be achievable on iPhone Safari.

Preferred production principles:

- low-to-moderate polygon counts;
- repeated-object instancing;
- shared materials;
- texture atlases where useful;
- LOD for vehicles/people/props;

- culling;
- hidden-floor suppression;
- bounded animated-agent counts;
- restrained transparency;
- restrained dynamic shadows;
- baked/simple lighting where appropriate;
- no shader complexity that exists only for close-up beauty.

The goal is to preserve the **visual language**, not reproduce concept-art render fidelity literally.

49.32 Accessibility and readability

Never encode important state using color alone.

Use combinations of:

- silhouette;
- icon;
- pattern;
- animation/state;
- label;
- color.

Ensure:

- key text remains legible at iPhone size;
- selected floor is obvious;
- interactable objects have sufficient visual contrast;
- important states survive common color-vision differences.

49.33 Explicit DO / DON'T

DO

- make a keypad slightly oversized so the player recognizes it;
- use a clean cutaway to expose interiors;
- animate carts, doors, elevators and staff truthfully;
- make loading and circulation readable;
- use wear to show property age;
- keep branding restrained;
- let small operational stories make the lot feel alive.

DON'T

- make the game photorealistic;
- turn people into oversized cartoon mascots;
- flatten the world into Prison Architect-like sprites;
- use neon/casino reward effects;
- make every surface glossy and pristine;
- hide gameplay under decoration;
- use concept-art beauty as an excuse to ignore iPhone performance;
- show actions that did not happen in the simulation.

49.34 Visual acceptance test

A representative gameplay screenshot should pass all of the following:

1. Immediately reads as contemporary American self-storage.
2. Property layout can be understood without opening a menu.
3. Drive-up vs interior storage is obvious.

4. Loading area is identifiable.
5. Carts are identifiable.
6. Elevator/stair access is identifiable in multi-story properties.
7. Customers and staff are visibly doing understandable work.
8. Important operating equipment can be recognized.
9. The scene remains readable at iPhone display size.
10. The property looks attractive enough that building and improving it feels rewarding.
11. The world does not look like a sterile architectural rendering.
12. The world does not look like a generic mobile-game cartoon.
13. Visual state does not contradict authoritative simulation state.

50. Audio, Sound, and Music Direction Bible

50.1 Canonical audio north star - LOCKED TARGET

Warm, tactile, grounded operational sound with playful tycoon-game character.

Real contemporary self-storage sounds - metal doors, carts, gates, keypads, elevators, HVAC, vehicles, tools, footsteps and customer activity - cleanly stylized so important actions remain recognizable without becoming noisy or hyper-realistic.

Music is calm, optimistic and lightly playful, supporting thoughtful building and management with no sense of time pressure.

Audio, animation, and visual state should reinforce the same authoritative event.

50.2 Audio design goals

Audio should make the property feel:

- physical;
- alive;
- understandable;
- calm;
- satisfying to operate.

It should not make the property feel:

- frantic;
- arcade-like;
- industrially deafening;
- constantly alerting;
- melodramatic.

50.3 The soundscape hierarchy

The soundscape has four layers:

1. Operational object sounds

Gate, keypad, door, cart, elevator, HVAC, tools.

2. Activity ambience

Footsteps, distant customer movement, office activity, vehicles, construction.

3. Environmental ambience

Wind, birds/suburban ambience, city bed, rain, evening ambience.

4. Music

Calm thematic score that supports long-form play.

The mix should leave room for important gameplay cues.

50.4 Spatial audio philosophy

Where technically practical, important world sounds should originate from their property location.

Examples:

- gate motor at the gate;
- keypad beep at the keypad;
- cart wheels at the cart;
- elevator chime at the elevator lobby;
- Tech tools at the repair location;
- HVAC hum near the plant;
- construction noise at the active project.

As the camera moves and zooms:

- nearby operational sounds become clearer;
- distant sounds blend into ambience;
- the player should never hear every object equally loudly.

The property should have an acoustic sense of place.

50.5 Signature operational sounds

Gate

- motor start;
- movement;
- stop/latch;
- fault/stall variant.

Keypad

- button/input tick if used;
- success tone;
- refusal/error tone;
- offline/fault cue.

Roll-up door

- latch;
- metal roll;
- stop;
- older-door rattle variants for aged properties.

Automatic/wide sliding door

- sensor/open;
- glide;
- close;
- fault/service state.

Carts

- rubber/urethane wheel roll;
- subtle metal rattle;
- threshold bump;

- empty vs loaded variation;
- damaged-wheel variant where appropriate.

Elevator

- arrival chime;
- door open/close;
- movement motor bed;
- button/call cue only if visible;
- service/fault cue.

HVAC

- low equipment hum;
- fan start/stop;
- stressed/faulted variation;
- never so loud that climate buildings become unpleasant.

Lighting/power

- restrained switch/relay confirmation;
- fault cue;
- avoid constant electrical buzz.

50.6 Vehicle audio

Vehicle classes should differ subtly:

- sedan;
- SUV;
- pickup;
- cargo van;
- box truck;
- moving truck;
- RV later.

Audio behaviors:

- arrival;
- idle;
- low-speed movement;
- reverse cue only when appropriate;
- door close;
- loading ambience.

Avoid making normal facility traffic sound like a racing game.

50.7 Customer audio

Customers do not need constant voiced chatter.

Use:

- footsteps;
- box handling;
- cart use;
- door interaction;
- restrained conversational murmur in occupied areas;
- occasional nonverbal effort/reaction only if stylistically appropriate.

Fully voiced dialogue is not required for core comprehension.

Conversation UI can use:

- subtle open/close cue;
- optional short vocal texture;
- text remains authoritative for meaning.

50.8 Staff audio

Staff actions should sound distinct:

- sweep/mop;
- cart recovery;
- make-ready;
- keypad repair;
- elevator service;
- HVAC service;
- camera/light maintenance;
- office interaction.

A player zoomed near a staff member should hear enough to understand that real work is happening.

50.9 Construction audio

Construction sound should track actual construction stage.

Examples:

- marking/site prep;
- light machinery;
- drilling/fastening;
- framing/shell work;
- door/equipment installation.

At wide zoom, active construction becomes a soft local sound bed.

At close zoom, detail increases.

Avoid a single looping "construction noise" for every project type.

50.10 Cleaning and maintenance sound

Cleaning:

- broom;
- mop;
- trash bag/bin;
- cart movement;
- restroom/service-room ambience where useful.

Maintenance:

- hand tools;
- drill/driver;
- panel open/close;
- metal impact;
- diagnostic beep;
- restrained electrical/tool effects.

The audio should reinforce what animation shows.

50.11 Environmental ambience by property type

Suburban

- distant traffic;

- birds/wind;
- occasional neighborhood ambience.

Urban

- denser traffic bed;
- distant siren/horn only very sparingly;
- building/mechanical ambience.

Large mixed/vehicle

- more vehicle movement;
- open-lot wind;
- larger-property spatial feel.

Interior

- HVAC;
- softened footsteps;
- cart reverberation;
- elevator sounds;
- distant doors.

Interior ambience should feel acoustically different from exterior without becoming cavernous.

50.12 Day/night ambience

Morning/day

- brighter outdoor ambience;
- more customer/vehicle activity.

Evening

- quieter bed;
- more audible equipment/lighting ambience;
- reduced office activity depending on hours.

Night

- restrained ambience;
- HVAC, distant traffic, gates, security activity;
- no horror-game tone.

50.13 Weather sound

Rain

- roof/canopy patter;
- wet tire movement;
- stronger covered-loading ambience.

Wind

- restrained fence/sign/canopy movement only where believable.

Weather should make infrastructure choices feel tangible without drowning the mix.

50.14 Gameplay feedback cues

UI/gameplay cues should be short, readable, and restrained.

Needed families:

- valid placement;
- invalid placement/refusal;
- construction committed;

- construction complete;
- commissioning complete;
- unit becomes rent-ready;
- lease/move-in confirmed;
- task assigned;
- task completed;
- equipment fault;
- equipment repaired;
- payment/rent collected;
- delinquency state change;
- staffing action;
- save confirmation/error;
- scenario objective progress/completion.

Ordinary events should not all sound equally important.

50.15 Alert hierarchy

Use three broad urgency bands:

Informational

- placement;
- task complete;
- lease;
- ordinary payment.

Attention

- maintenance due;
- cart shortage;
- service queue;
- equipment degradation.

Critical

- blocked access;
- elevator outage affecting floors;
- severe HVAC failure;
- major financial warning.

Critical sounds must remain rare enough to retain meaning.

50.16 Music identity

Music should be:

- warm;
- optimistic;
- lightly playful;
- modern;
- unobtrusive;
- loop-friendly;
- suitable for long management sessions.

Instrumentation can lean toward:

- light percussion;
- bass;
- muted guitar;
- soft keys;

- marimba/mallet textures used carefully;
- gentle synth pads;
- subtle acoustic textures.

Avoid:

- bombastic orchestral scoring;
- high-BPM arcade music;
- constant corporate ukulele;
- slapstick comedy cues;
- tense "danger" music for ordinary business problems.

50.17 Music modes

The score can use related arrangements rather than completely different soundtracks.

Potential states:

- title/menu;
- planning/build mode;
- normal daytime operation;
- evening/night operation;
- milestone/expansion;
- scenario completion;
- company/portfolio view.

Transitions should be soft.

The music should not change dramatically because occupancy moved by a small percentage.

50.18 Adaptive music

Adaptive behavior should be subtle.

Good:

- lower intensity while paused/inspecting;
- slightly fuller arrangement when major construction opens;
- gentle milestone flourish;
- reduce musical density when close to rich operational sound.

Avoid:

- "bad economy" danger music;
- constant stingers for every lease;
- tension systems that imply time pressure.

50.19 Silence and breathing room

Silence is part of the design.

A small property on a quiet morning can be mostly:

- distant traffic;
- HVAC;
- occasional gate;
- cart;
- footsteps;
- light music.

Do not fill every second with sound.

50.20 Music and sound mixing priorities

Priority:

1. critical player feedback;
2. selected/nearby operational sound;
3. dialogue/conversation cue if active;
4. general ambience;
5. music.

When many sounds compete:

- attenuate distant repeated sounds;
- limit simultaneous identical events;
- duck music slightly for important cues;
- aggregate dense activity into ambience.

50.21 UI sound language

UI should sound:

- clean;
- tactile;
- modern;
- understated.

Build mode:

- soft snap;
- confirm;
- invalid/refusal.

Panels:

- restrained open/close;
- tab switch.

Time controls:

- distinct Pause / 1x / 2x / 4x confirmation;
- never a stressful ticking clock.

Money:

- do not use slot-machine coin cascades.

50.22 Scenario audio

Scenarios can have light identity through:

- property ambience;
- music arrangement;
- weather;
- equipment age;
- urban/suburban soundscape.

Do not create a one-off soundtrack for every scenario unless production scale later justifies it.

50.23 Audio accessibility and settings

Player controls should include:

- Master Volume;
- Music;

- Sound Effects;
- Ambience;
- Voice/Dialogue if voice is ever added.

Support:

- full comprehension with audio muted;
- no required information delivered only through sound;
- optional vibration/haptics only where platform support and product direction justify them.

50.24 Mobile Safari audio lifecycle

Audio design must respect browser/mobile restrictions.

Target behavior:

- audio context begins suspended;
- unlock occurs only after a valid user gesture;
- no autoplay music before interaction;
- pre-unlock sounds are discarded rather than building a stale queue;
- interrupted iOS audio contexts are treated defensively and rebuilt if required;
- returning from background must not replay old one-shot events;
- audio failure must never block gameplay.

50.25 Audio authority boundary

Audio is a downstream consumer of authoritative gameplay events.

Audio may:

- play a cue after a confirmed event;
- spatialize an operating object;
- vary presentation based on authoritative state.

Audio may not:

- decide whether an action succeeds;
- trigger gameplay state;
- create a fake lease, payment, task, repair, alert, or incident;
- infer hidden gameplay rules independently from presentation.

If the simulation says the elevator did not arrive, the audio layer does not invent an arrival chime.

50.26 Audio asset families

Production asset families should include:

World

- gates;
- doors;
- elevators;
- carts;
- HVAC;
- lights/power;
- security/keypads;
- construction;
- cleaning;
- maintenance;
- vehicles.

Character

- footsteps;
- handling/boxes;
- restrained exertion;
- staff task sounds.

Environment

- suburban;
- urban;
- interior;
- day;
- night;
- rain/wind.

UI

- build;
- operate;
- business;
- growth;
- time controls;
- save;
- warning;
- completion.

Music

- menu;
- build;
- operate;
- evening;
- company;
- milestone.

50.27 Audio DO / DON'T

DO

- make cart wheels recognizable;
- let gates/keypads/elevators have signature sounds;
- localize world sounds;
- keep quiet properties quiet;
- make maintenance sound like the work shown;
- use music to support focus and comfort;
- make day/night and interior/exterior acoustically distinct.

DON'T

- create a wall of constant beeps;
- use cash-register sounds for every revenue event;
- use alarm sirens for minor issues;
- use music to manufacture urgency;
- autoplay before a user gesture;
- replay stale sounds after resume;
- make audio a second gameplay authority;
- require sound to understand a problem.

50.28 Audio acceptance test

A representative play session should pass:

1. Muted play remains fully understandable.
2. With audio on, the property feels substantially more alive.
3. Gate, keypad, cart, elevator, loading and maintenance sounds are distinguishable.
4. The player can often infer nearby activity by sound without opening a panel.
5. Dense activity does not become noise.
6. Music supports long play without creating pressure.
7. Interior and exterior spaces feel acoustically different.
8. Day/night/weather produce subtle atmosphere changes.
9. Important alerts remain rare and meaningful.
10. Audio never contradicts gameplay truth.
11. Background/resume does not replay stale events.
12. Audio failure does not affect gameplay correctness.

PART XIX - TECHNICAL AND INFORMATION ARCHITECTURE

51. Technical Authority Boundary

Authoritative gameplay belongs in the simulation layer. Presentation:

- captures input;
- requests authoritative actions;
- renders authoritative state;
- formats explanations;
- animates confirmed results.

The browser/UI must not independently decide:

- build validity;
- prices/economy;
- routing results;
- task duration;
- elevator capacity outcome;
- cart availability outcome;
- security/coverage effects;
- save semantics.

52. Required Simulation Domains

Long-term domains:

- parcel/grid;
- building/floor geometry;
- vehicle routing;
- pedestrian/cart routing;
- units/product inventory;
- construction/commissioning;
- customer agents;
- carts;
- elevators/vertical connectors;
- staff/actors/tasks;
- maintenance/condition;
- security/access;
- lighting;
- HVAC/power/water abstraction;
- leasing/pricing/demand;
- billing/economy;
- delinquency/resolution;
- progression/company;
- persistence.

They should be introduced incrementally, but the architecture should avoid trapping the game in a drive-up-only model.

Target domains:

- parcel/grid/floors;
- construction;

- buildings/rooms;
- access/routing;
- vehicles;
- pedestrian/cart movement;
- units;
- leases;
- customers;
- market/demand/pricing;
- actors/staff/tasks;
- carts;
- elevators;
- utilities;
- security;
- condition/maintenance;
- cleaning;
- economy/ledger/capital;
- progression/scenarios/company;
- incidents;
- save/resume.

Each authoritative domain should expose enough state for presentation and diagnostics without duplicating gameplay rules in the UI.

53. Save / Resume

The player can stop at any time, leave Safari/app, and return to the same authoritative facility state.

Persistent gameplay eventually includes:

- parcel/buildings/floors;
- units;
- doors/access systems;
- carts and corral assignment if they have persistent location/state;
- elevators/equipment condition;
- staff/tasks;
- construction;
- leases/economy;
- progression/tutorial state.

Presentation-only animation does not become gameplay authority merely because it looks alive.

Save/resume is a product requirement, not a convenience feature.

The game must persist every durable authoritative system that can affect continuation.

54. Developer Diagnostics and Proof

A complex simulation needs tools that let development verify causality.

68.1 Player diagnostics

Every major causal system should have:

- physical tell;
- inspection explanation;
- focused overlay;
- business summary where relevant.

68.2 Developer diagnostics

Debug builds should be able to inspect:

- route chosen;
- route length;
- unit state;
- lease state;
- task queue;
- actor assignment;
- cart location/state;
- elevator queue/dispatch;
- utility capacity/coverage;
- customer decision inputs;
- ledger entries.

Developer diagnostics do not need to be player-facing.

68.3 Deterministic tests

Core simulation tests should use fixed seeds and action scripts where randomness exists.

For any new spatial mechanic, a useful test pair is:

- compact/control layout;
- deliberately inefficient layout;
- same market, seed, inventory, and actions.

The result must show whether geometry actually changed operations.

68.4 No fake causality

If an overlay says "elevator delay reduced leasing," that statement must be backed by authoritative coupling.

Presentation may summarize truth; it may not create truth.

PART XX - CAUSAL DESIGN

55. Complete Causal Matrix

The following matrix defines what the game is trying to make legible.

Player choice	Operational result	Economic result	Visible evidence
More/closer carts	less search/wait/recovery burden	better convenience; potentially stronger leasing/retention when coupled	full corral, customers get carts quickly
Better corral placement	shorter loading/elevator transfer	less friction/labor	cart routes visibly shorter
Wider/loading door	more entrance throughput	better interior usability	shorter doorway queue
Covered loading	weather-protected loading + organized staging	quality/convenience value	customers unload under canopy
Elevator capacity	more vertical throughput	upper-floor product performs better when demand exists	lobby queue falls
Second elevator	throughput + redundancy	higher capex/maintenance but more usable vertical density	parallel elevator service
Wider hallway	lower movement friction	sacrifices rentable area for usability	fewer cart conflicts
Better lighting	stronger comfort/security coverage	operating cost + quality effect	dark areas disappear
Camera placement	coverage improvement	security-quality effect	blind spots disappear
Better keypad/gate	faster/reliable access	lower access friction, maintenance cost	vehicle/building queue improves
HVAC capacity	viable climate area	higher rent opportunity + higher OpEx	climate zone remains in spec
More Tech capacity	faster equipment repair	less downtime + more payroll	service queue shrinks
More Porter capacity	cleaner property, better cart recovery	quality/retention + payroll	fewer dirty areas/misplaced carts
Better office placement	easier customer service/staff origin later	service efficiency	shorter service routes
Higher rent	higher revenue per lease, lower move-in response	occupancy/revenue tradeoff	demand/lease response
Expansion	more capacity	capex + future revenue opportunity	new physical wing/units
Poor expansion layout	long routes/congestion	higher labor/friction	queues, travel, backlog
Preventive maintenance	fewer severe failures	planned service cost vs avoided downtime	fewer emergency faults
Manager delegation	less direct player task handling	payroll vs scalable operations	property continues operating under policy

The exact economic magnitude of several rows remains subject to implementation and playtesting. The causal direction must be explicit before the UI claims it.

56. Feature Definition of Done

A feature is not complete because:

- a button exists;
- an icon exists;
- a number changes;
- an object can be placed.

A feature is complete when:

1. The player has a reason to care.
2. The player can use it on the property.
3. The simulation responds authoritatively.

4. The response is visible.
5. The business consequence is truthful where one exists.
6. A poor outcome can be diagnosed.
7. The player has at least one meaningful response.
8. Save/resume preserves it correctly.
9. It works at the iPhone interaction/readability target.
10. It fits the broader construction -> operation -> economics -> growth loop.

A feature is not complete because it has a button or object. It is complete when the player can understand its purpose, use it, see its authoritative consequence, recover from poor performance, and save/resume it reliably.

57. Signature-System Acceptance Tests

45.1 Construction grammar

PASS when a new player can build a simple unit row without guessing what frontage/access it needs, and the result becomes a real part of routing/economics.

45.2 Interior storage

PASS when an interior unit is mechanically different from a drive-up unit because the customer actually uses loading, doors, hallways, carts and vertical access where relevant.

45.3 Carts

PASS when:

- carts physically exist or are represented faithfully;
- customers claim/use them;
- scarcity is visible;
- cart location matters;
- misplaced carts create recoverable work;
- changing cart quantity/corral placement changes customer/staff behavior.

45.4 Elevators

PASS when:

- upper-floor customers use them;
- carts consume capacity;
- queues form visibly;
- placement/capacity changes outcome;
- maintenance can degrade service;
- the player can recover through service/upgrade/layout.

45.5 Covered loading

PASS when the dock changes the customer route/capacity/convenience rather than only adding a roof prop.

45.6 Keypads

PASS when customers use them, failure/slow service is visible, and repair/upgrade affects access.

45.7 HVAC

PASS when climate inventory requires real capacity/coverage and an HVAC service issue creates visible operational consequences.

45.8 Amenities

PASS when restrooms/water fountains improve authenticity/quality and create light operational work without turning into needs micromanagement.

Additional acceptance tests:

Empty-lot start

PASS if a new player can move from raw parcel to open facility without discovering an unstated mandatory requirement after spending their capital.

Scenario

PASS if completion/failure conditions are based on actual authoritative state and the scenario supports at least one coherent player-driven solution rather than a scripted sequence.

Sandbox

PASS if the player can build and operate indefinitely, and Creative settings relax restrictions without breaking the simulation.

Multi-property

PASS if an unattended managed facility continues routine operations correctly and surfaces exceptions without requiring constant manual babysitting.

PART XXI - SCOPE AND PRODUCTION

58. Production Sequence

This is a build sequence, not a claim of current authorization.

Phase A - Preserve trustworthy foundations

Do not regress:

- save/resume;
- economy truth;
- lease/turnover;
- time controls;
- current construction/access authority;
- C++ gameplay authority.

Phase B - Construction grammar proof

Implement the smallest version of:

- building shell;
- hallway;
- interior unit row;
- door orientation;
- commissioning.

Goal: prove that drawing a small interior storage wing is satisfying on iPhone.

Phase C - Interior logistics proof

Add:

- loading area;
- wide sliding door;
- carts;
- cart corral;
- interior customer route;
- cart recovery.

Goal: prove drive-up and interior storage feel mechanically different.

Phase D - Vertical logistics proof

Add:

- floor model;
- stairs;
- elevator;
- capacity/queue;
- upper-floor unit routing.

Goal: prove a second floor creates fun design tradeoffs.

Phase E - Building systems proof

Add:

- lighting;
- cameras;

- building keypad;
- power-capacity abstraction;
- HVAC/coverage;
- climate units;
- restroom/water service.

Goal: prove a premium indoor facility is a connected operating system.

Phase F - Operations depth

Deepen:

- cleaning;
- maintenance;
- Porter/Tech;
- area assignments/priorities;
- customer conversations;
- preventive maintenance.

Phase G - Tutorial integration

Rewrite Maple Street around the proven systems so the player learns them organically.

Phase H - Company growth

Add:

- property Manager;
- multiple properties;
- acquisition/development choices;
- capital allocation;
- portfolio comparison.

The sequence should now be interpreted against this Master GDD:

1. protect save/economy/authority;
2. prove the property-as-game-board thesis;
3. build interior construction grammar;
4. build customer logistics (loading/carts);
5. build vertical logistics (floors/elevators);
6. build infrastructure/security/amenities;
7. deepen staff/maintenance/customer interaction;
8. integrate tutorial/campaign;
9. author scenarios;
10. enable robust sandbox;
11. build company/multi-property layer.

Each phase must result in playable game value, not just backend capability.

59. Scope Protection

Do **not** interpret this GDD as permission to build every listed system simultaneously.

The target vision is broad. Implementation remains bounded.

Before a subsystem expands, it must prove:

- it changes a player decision;

- it produces a visible result;
- it connects to operations/economics;
- it remains understandable on iPhone;
- it does not create more maintenance burden than fun.

60. Deliberately Not Simulated

Clarity also requires saying what the game refuses to become.

Do not model at high fidelity unless later evidence proves it fun:

- individual electrical wires;
- individual water pipes;
- duct routing;
- structural engineering calculations;
- fire-code paperwork;
- detailed permits;
- tax accounting;
- insurance-document processing;
- legal lien paperwork;
- employee HR administration;
- individual customer inventory item physics;
- manual elevator dispatch;
- manual cart dragging as a routine management task;
- detailed vehicle parking maneuvers;
- realistic janitorial chemical inventory;
- per-customer thirst/bladder meters;
- construction-trade staffing;
- fully modeled weather disasters.

The simulation should be deep in **decisions and consequences**, not deep in paperwork.

61. Open Tuning Questions

Leave these to experiments/playtesting:

- exact hallway width values;
- cart return probability;
- cart costs/counts;
- elevator capacity/speed;
- how strongly convenience affects leasing;
- exact loading-bay throughput;
- exact HVAC capacity costs;
- exact maintenance wear rates;
- staff walking speed;
- upper-floor price differences;
- exact construction durations;
- how much water fountains affect quality;
- exact customer-dialogue frequency;
- demolition/move timing and refunds;
- multi-property time model.

The GDD should define **what the choice means**, not lock arbitrary numbers prematurely.

These are intentionally not fixed in the master design because they need implementation and playtesting:

- exact construction costs and times;

- hallway capacity coefficients;
- cart purchase cost and ideal cart counts;
- elevator capacity/speed/maintenance rates;
- utility operating costs;
- customer sensitivity weights;
- staff wages and work rates;
- maintenance wear rates;
- scenario target values;
- financing rates/limits;
- exact property acquisition prices;
- specific market demand curves.

PART XXII - FINAL PRODUCT DEFINITION

62. What the Player Should Be Able to Say

After the tutorial:

- "I know how a unit becomes rentable."
- "I know why layout matters."
- "I know how customers get into and through the facility."
- "I know what carts are doing and why I care."
- "I know where my money comes from and where it goes."
- "I know why I would hire someone."
- "I know how to expand without being rushed."

After the multi-story property:

- "I know how loading, carts, elevators and hallways work together."
- "I can diagnose why an upper-floor wing is inconvenient."
- "I can decide whether to spend money on capacity, staff, maintenance or a redesign."

After the company stage:

- "I can look at a property and understand what kind of investment it needs."
- "I don't have to personally do every task anymore."
- "My company grew because I learned how to design and operate good storage properties."

63. Final Master Vision

The player is not selecting abstract storage capacity from a menu.

The player owns land and turns it into a functioning self-storage facility.

They decide where vehicles enter, where customers park, where the office goes, how drive aisles move traffic, where loading happens, how an indoor building is arranged, where carts wait, how customers reach upper floors, what security covers each area, how HVAC supports climate storage, and where staff work happens.

Then the simulation actually uses those choices.

A tenant can drive through a gate, enter a code, park beneath a covered loading area, discover whether a cart is available, move through a wide sliding door, wait for an elevator, travel down a hallway, and reach a unit. Another customer may simply pull directly to a drive-up door. A Clerk can handle a code issue in the office. A Porter can recover carts from the second floor. A Tech can repair an elevator or keypad. The Owner can still step in personally.

The player's layout creates operational consequences.

Operations create costs, labor, customer experience, occupancy pressure, revenue, and cash.

Those economics determine what the player can improve or build next.

That growth creates a larger and more complex property, which creates new operating decisions.

The player is allowed to build imperfectly. A weak decision becomes a readable management problem, not an invisible punishment. Pause exists so the player can study it. Diagnostics explain it. Management and construction provide ways to fix it.

The tutorial teaches this on one understandable property.

Scenarios deliberately challenge different parts of the skill set.

Sandbox lets the player build and operate freely for as long as they want.

The company layer lets the player take what they learned and apply it across multiple properties and markets.

The final fantasy is not merely:

****"I own storage units."***

It is:

****"I know how to design and operate a good self-storage property because I have watched my own facilities work, fail, recover, improve, and grow."***

That is Self Storage Tycoon.

Document Governance Note

This Master GDD exists to prevent product drift. It defines the target game, including presentation. It does not replace source control. Implementation claims must be verified against live GitHub, and unresolved tuning values should remain explicit until implementation and playtesting justify them.